

Sports Medicine and Adaptive Sports

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SUMMARY PEARLS

- Physiatrists are primed to be leading experts in caring for adaptive athletes; understanding pathology in this patient population is critical.
- Understanding natural healing processes and management principles of treating acute athletic injuries continues to evolve and is described in detail in this chapter.
- Updates on select injuries and illness encountered by the sports medicine physician are detailed in this chapter, including critical updates in concussion, management of cervical spine injuries, and sudden cardiac death in athletes.
- Care of the transgender athlete has become a prominent and important source of discussion in the sports medicine community over the past several years.

Introduction

Physiatrists are uniquely qualified to serve as sports medicine physicians and care for the recreational and competitive athlete and those who exercise for health-related benefits. Physiatrists are trained extensively in musculoskeletal medicine and injury, functional rehabilitation, and to coordinate and lead a team of professionals to optimize care of patients. The sports medicine team includes the athlete, the athlete's family, specialty and primary care physicians, athletic trainers, physical and massage therapists, chiropractors, dietitians, psychologists, and coaches. Physiatrists are also skilled in prehabilitation, or preventive rehabilitation, which is a critical aspect of care for any athlete involved in routine physical training. Exercise physiology, emergency medical care, and general medical care are also important to any well-rounded sports medicine physician. This chapter covers the breadth of sports medicine for the physiatrist, including the general role and medicolegal aspects of being a team physician, sporting event administration with a particular focus on emergency preparedness, athletic conditioning and training principles, injury prevention and functional rehabilitation, biomechanics of sports, pharmacology in sports, emergency assessment and care of the athlete, common medical and neurological conditions in athletes, and a review of specific populations of athletes and their common ailments, including a special focus on athletes with disabilities and adaptive sports medicine. This chapter is not intended to be a musculoskeletal medicine chapter because other chapters in this text cover in-depth musculoskeletal issues that overlap with the musculoskeletal injuries seen in athletes.

Role of the Team Physician

The team physician has multiple overlapping roles within the framework of the team itself, the individual athletes and their families, the school or overseeing organization, and the community. The primary responsibility is caring for the health and well-being of the individual athlete. The chief duties are to determine initial medical eligibility, provide care for the injured athlete, facilitate and determine return to play (RTP), create and maintain emergency preparedness, oversee the health of the team's overall training program, supervise the personnel providing health care for the team, and protect against institutional and personal liability.¹⁸³ The team physician is responsible for conveying information to other medical specialists, allied health professionals, athletic trainers, and coaches as well as for creating and coordinating health care plans for the athletes.¹⁸

The primary obligation of the sports medicine physician is always to the athlete as patient. The specific details of confidentiality and the flow of information are defined by the Health Insurance Portability and Accountability Act. Specifically, the components of consent for treatment and the authorization to release information are most pertinent to the team physician. It is prudent to review this document and the applicable state law and to discuss these matters with all involved parties (e.g., athletes, coaches, families) before issues arise. Beyond the medical obligation to the athlete, the team physician has an assumed medicolegal responsibility. Physicians receiving financial or other obvious remuneration for services are bound by normal medicolegal responsibilities. The physician might be held legally liable if injury

occurs as a result of negligence or treatment below the professional standard of care. Those physicians who volunteer their services may not be covered by Good Samaritan laws, particularly if they are the “designated” medical provider for an event, regardless of whether they are remunerated for their services. Preparticipation physical evaluations (PPEs) are not covered under Good Samaritan laws. The team physician is also responsible for the final decision regarding clearance to play or RTP after injury. These issues can be complex, and several fundamentals must be met before the athlete should be cleared. When recovering from an injury, the athlete must have completed appropriate rehabilitation, a health care professional must document the athlete’s recovery, and the risks of return must be discussed and documented. The team physician must obtain the appropriate consultant opinions when dealing with conditions or injuries outside the scope of practice. Although some states have determined that athletic participation implies assumed risk, this is not universal. Team physicians have been held liable for injury occurring after full RTP has been granted. In cases of differing opinions of the team physician and a consultant, the final responsibility typically lies with the team physician.

Historically, the traveling team physician presented a unique issue in the United States given that licensure and malpractice coverage are state specific. However, in October 2018, the Sports Medicine Licensure Clarity Act was signed into law, allowing a team physician providing medical services to an athlete or team staff member to be covered under the physician’s home state’s licensure while across state lines.⁹³

Event Administration

The two most common competition venues where the sports medicine physician practices are on the sidelines of a sporting event (e.g., football, soccer, gymnastics) and at mass participation endurance events (e.g., marathons, road races, triathlons). These are unique venues that require appropriate planning, organization, rehearsal, and communication to provide optimal medical care. Both require significant preseason or pre-event planning.¹⁰¹ From an administrative perspective, it is essential that the team physician or chief medical officer develops a chain of command that defines the responsibilities of the medical providers, emergency medical services, event officials, and other parties involved. Establishing and regularly rehearsing an emergency action plan (EAP) is critical. Reviewing the medical equipment that will be available at the event and the protocols for attending to an injured athlete is a standard component of preparation. The team physician must also assess environmental concerns and playing conditions and have a policy to modify or suspend practice or competition if adverse conditions exist.

Administering care at a mass participation endurance event such as a marathon requires unique planning given the sheer number of participants, the unique medical illnesses encountered, the length of the course, the number of medical sites needed, and the large medical team required to staff such an event. Given its size, the potential impact of the event on the community and its resources needs to be accounted for in planning and implementation. These large mass participation events require the coordination of public, private, and medical agencies, as well as effective communication among them during race planning and race day. Although any marathon has a predictable number of people requiring medical assistance (2%–20% of participants, depending on event size, duration, and weather conditions), if these numbers

should increase unexpectedly, there is a potential for the community resources to become overwhelmed if they are not allocated appropriately.^{39,196} For this reason, communication among the medical team, race officials, emergency medical services, local hospitals, and participants is crucial to help prevent and respond to incidents during the race. It is the role of the chief medical officer to coordinate these effective lines of communication.

Medical staff need to be familiar with the more common ailments that athletes can present with during or after the event and to have the appropriate equipment available in advance. For example, exercise-associated collapse (EAC), a common medical entity seen in marathon medical tents, has a broad differential diagnosis, which includes benign to potentially life-threatening disorders (see later). Appropriate medical equipment for a marathon includes devices for rapid electrolyte blood testing, cardiac monitoring capabilities, automated external defibrillators (AEDs), thermometers, rapid warming and cooling protocols and equipment, and personnel and equipment for starting intravenous therapy. Prevention of medical tent encounters is an even better way to enhance safety. Preventing race day maladies in marathoners requires educating racers and properly coordinating appropriate resources (aid and fluid replacement stations) along the course. It is important to educate the runners in the months, weeks, and even the day before and day of the event to “listen to their bodies,” train adequately, determine/estimate fluid needs before the race and stick to a plan during the race, and adjust race/time expectations if the weather conditions are adverse. Education is an important strategy to lessen morbidity (and potentially mortality) during and after the race.

Principles of Conditioning and Training

Effective training programs require a fundamental understanding of strength, flexibility, and endurance development, as well as the basic principles of specificity, individuality, periodization, overload, and tapering. Please refer to Chapter 16 for an overview of strength, flexibility, and endurance training. The body’s adaptation to physical training is metabolically and neuromuscularly specific to the exercise performed. The principle of sport specificity highlights the importance of training in the same sport in which the athlete will be competing. For example, although cycling and cross-country skiing are valuable forms of aerobic exercise, an athlete who is training to run a marathon needs to focus on running as the major component of training to ultimately maximize running performance. However, it must be realized that individuals respond differently to the same training stimulus. For example, a less-conditioned athlete will make larger gains in fitness than a more conditioned athlete who embarks on a training program. The principle of overload requires that the training stimulus be greater than how the athlete normally performs in competition. Exercise frequency, duration, and intensity are manipulated to produce overload. These specific training variables (frequency, duration, and intensity) must be periodically increased for an athlete to progress (Box 40.1).

• BOX 40.1 Training Program Design Variables: FITT

- **F:** Frequency of training
- **I:** Intensity of training
- **T:** Type of exercise mode
- **T:** Time of exercise duration

	A 12-month plan																								
Macro-cycles	Preseason								Competitive								Postseason								
Meso-cycles																									
Micro-cycles																									

• **Fig. 40.1** A template for a periodized training program. (From Bachtel N, Baron R, Smekal G. Principles of exercise physiology and conditioning. In: Frontera WR, Herring SA, Micheli LJ, et al., eds. *Clinical Sports Medicine: Medical Management and Rehabilitation*. Saunders; 2007.)

Periodization

Periodization is one structured training approach that highlights the concept of training variables. It was developed by a Russian physiologist in the 1960s but has had a significant resurgence in current training programs (Fig. 40.1). In periodization, training is divided into defined “periods” to allow buildup of training stresses, time for rest and adaptation to training, and continual progression of fitness. In linear periodization, a year of training is broken down into macrocycles (commonly lasting multiple months), which are divided into shorter mesocycles (commonly lasting 1 month), which are then subdivided into microcycles (commonly lasting 1 week). Microcycles are generally the weekly training programs. Each mesocycle can be made up of 4 weekly microcycles, with a gradual buildup of training frequency and intensity over the first 3 weeks and then a slight decrease in the fourth week to allow for rest and the subsequent metabolic training adaptations to occur. The athlete is then ready to enter the next 4-week mesocycle and progress through another 4 weekly microcycles. The amount and intensity of training in each mesocycle depend on where the athlete is in relationship to the competitive season, which is how macrocycles are structured.

A typical yearlong training program will have three macrocycles: preseason or “buildup phase,” competitive season or “maintenance and fine-tuning phase,” and postseason or “recovery phase.” The preseason macrocycle is typically the longest, occurring before the competitive season. It is designed to develop fitness in anticipation of the more intense training to follow.²³⁶ During the preseason macrocycle, the athlete generally focuses on higher volume and lower intensity exercise.

The competitive season macrocycle follows the preseason period. The focus here is to develop and maintain peak fitness with a focus on high-intensity training and sport-specific technique drills.²³⁶ Because of the higher intensity training, the volume must be significantly reduced. Finally, the postseason macrocycle is a time for the athlete to recover from the previous year’s training and physically and mentally prepare for the next year of training and competition.²³⁶ The early portion of this period necessitates “active rest.” The athlete generally participates in unstructured, non-sport-specific recreational activities to allow for recovery from the stresses of the competitive season. This is an important period for the athlete to adequately recover from injuries, prevent overtraining, and take a “mental break” from the competitive season. During the competitive season, a typical microcycle includes a

tapering period just before competition. The taper is a short period of reduced training before an important competition, with the goal of optimizing performance. Although it is well recognized that a reduction in training before competition improves performance, it is not definitively known what is the most ideal taper strategy regarding altering training volume, intensity, and frequency.^{9,106,165} Recent metaanalyses demonstrate that the most efficient strategy to maximize performance gains is a 3-week or less taper during which training volume, but neither training intensity nor frequency, is progressively reduced by 41% to 60%.^{21,232} Tapering strategies are associated with a mean competition performance improvement of approximately 3% (range 0.5%–6%).¹³²

A second type of periodization is known as undulating periodization, which involves more frequent variations in intensity and volume and is often divided into biweekly, weekly, and daily periods. There does not appear to be a significant advantage of one type of periodization over the other in terms of strength gains for trained individuals; however, in untrained individuals, recent evidence suggests that undulating periodization may result in improved strength gains compared with linear periodization.^{98,161}

Overtraining Syndrome

Athletes, coaches, and sports medicine physicians should be able to recognize symptoms of overtraining. A short-term decrease in performance of 2 weeks or less that is followed by supercompensation is known as overreaching and is a mechanism to improve athletic performance. Nonfunctional overreaching occurs when the performance decrement lasts longer, at least 3 to 4 weeks, and is not followed by supercompensation. Overtraining syndrome occurs when the performance decrement lasts more than 3 to 4 weeks and is not followed by improvement in performance. Common symptoms of overtraining syndrome, in addition to a performance decrement, include generalized fatigue, mood disturbance, poor sleep, and increased rates of illness and injury.¹⁵⁶ This can have long-term detrimental effects on the athlete’s health and performance. The treatment is rest, from weeks to months, with gradual resumption of training. However, prevention is the best treatment, and following a periodized training program is one method to ensure adequate rest from more intense bouts of training.

Altitude Training

Many elite endurance athletes incorporate altitude training into their typical training program in hope of enhancing performance in competition. For those athletes who do not live in mountainous areas, there are a variety of “altitude tents” on the market that can be used to simulate living at high altitude. In 2006, the World Anti-Doping Agency (WADA) considered placing “artificially induced hypoxic conditions” on its Prohibited List of Substances/Methods, but ultimately left it off.²⁴⁴ The proposed physiological mechanism(s) responsible for the effect of altitude training on sea-level performance include accelerated erythropoiesis (increased erythrocyte volume) as the primary hematological effect, as well as nonhematological factors such as improved muscle efficiency at the mitochondrial level, glucose transport alterations, and enhanced muscle-buffering capacity via pH regulation.⁸⁵ Most scientists agree that the most effective form of altitude/hypoxic training for elite athletes is the “live high–train low” method whereby athletes “live high” to stimulate erythropoietin (EPO) and subsequently increase erythrocyte volume and “train low” at a higher intensity with improved oxygen flux

to induce beneficial metabolic and neuromuscular training adaptations.²⁴⁵ The optimal “dose” of altitude training is not known. Research suggests that for athletes to derive the most beneficial hematological effects while using natural altitude, they need to live at an elevation of 2000 to 2500 m for 14 to 22 hours a day for 19 to 28 days.¹¹⁹ For those using simulated altitude environments, 12 to 16 hours of hypoxic exposure might be necessary but at a higher elevation (2500–3000 m).²⁴⁵ Though popular among elite athletes, the true performance benefit of altitude training is largely unknown and minimal and temporary at best.¹¹⁹

Injury Prevention and Rehabilitation

Injuries that occur in the course of an athlete's life will be related to a multitude of causes. These include acute trauma, overuse, and injuries secondary to poor technique. Certain injuries are predictable and even preventable in many cases.

Kinetic Chain Assessment

The kinetic chain model is based on the idea that each complex, athletic movement is the summation of its constituent parts. For example, a quarterback's throwing motion is created through the action of feet on the playing surface, loading of the lower limbs, rotation of the hips and abdominal muscles, activation of the latissimus dorsi, stabilization of the glenohumeral joint by the periscapular muscles, loading of the throwing arm through the deltoid and biceps, and upper limb motion of elbow extension and wrist flexion. The upper limb acts as a “funnel” for the energy generated by the core and lower limbs.¹²⁰ Each of the “links” in the chain must function well to optimize performance and minimize potential tissue trauma.

“Catchup” occurs when an athlete tries to compensate with one segment for a deficiency in a separate segment. This phenomenon puts higher stress on the tissues of the distal segment and predisposes it to injury.¹²⁰ For example, a baseball pitcher might present with elbow pain for a variety of reasons. It could simply be an overtraining phenomenon or it might be caused by poor shoulder or hip range of motion (ROM), abnormal scapular biomechanics, or poor core strength and control.²⁵⁴ Relief can be achieved through temporary rest, but the pain will likely return unless the causative issue is corrected. The experienced sports medicine physician will identify the biomechanical deficit and correct the predisposing factors. It is important that the physician is able to explain these concepts to the athlete and the treatment team to achieve optimal results.

Rather than view all similar overuse injuries as the same issue, the sports medicine physiatrist is able to break down the complex motions of the athlete into the constituent parts, identify the maladaptive pattern, and create a rehabilitation program that will prevent the injury from recurring. When examining the injured athlete, it is vital to make a biomechanical diagnosis rather than to focus exclusively on the injured tissue. Although it is easy to focus exclusively on the painful tissue, it is essential to assess the entire athlete and the athlete's mechanics to make appropriate tissue and biomechanical diagnoses (Table 40.1).

Evaluating an injured athlete involves a thorough discussion of the current problem, previous athletic injuries in the same area, and other seemingly unrelated injuries. Discussion of training patterns, including intensity and volume, as well as competition details is important. Proper equipment fit and technique should

TABLE 40.1 Tissue Diagnosis and Biomechanical Diagnosis

Tissue Diagnosis	Potential Biomechanical Diagnosis
Lateral/medial epicondylitis	Posterior deltoid weakness
Hamstring strain	Overly tight hamstring, weak gluteal musculature
Metatarsal stress fracture	Supinated foot
Athletic pubalgia	Weak core musculature, tight hip girdle
Shoulder impingement with rotator cuff strain	Periscapular weakness or inhibition
Patellofemoral syndrome—patellar cartilage irritation or chondromalacia	Quadriceps and gluteal weakness or inhibition, overpronation
Repetitive ankle sprains—anterior talofibular ligament laxity	Weak peroneals, proprioceptive dysfunction

also be assessed. Differentiating between a slowly worsening insidious process and a more acute injury is key to creating an appropriate treatment plan. These subtle distinctions can frequently reveal the underlying cause of the current injury.

The physical examination of the athlete must be focused on finding the biomechanical culprit, rather than focusing solely on the painful tissue. The athlete must be examined in detail to fully assess the possible causes of the injury. Select upper limb tests that help to determine the role of the scapula in upper limb injuries include the scapular assistance test and the scapular retraction test.¹²² Core muscle function must be assessed dynamically and addressed as part of the rehabilitation plan.¹²³ Examples of physical examination maneuvers to evaluate core muscular function in the athlete are demonstrated in Figs. 40.2 to 40.5. The kinetic chain assessment is designed to assess flexibility, strength, and functionality of the affected limb. The examination essentially assesses multiplane sports-specific dynamics rather than single-plane, solitary joint movements.

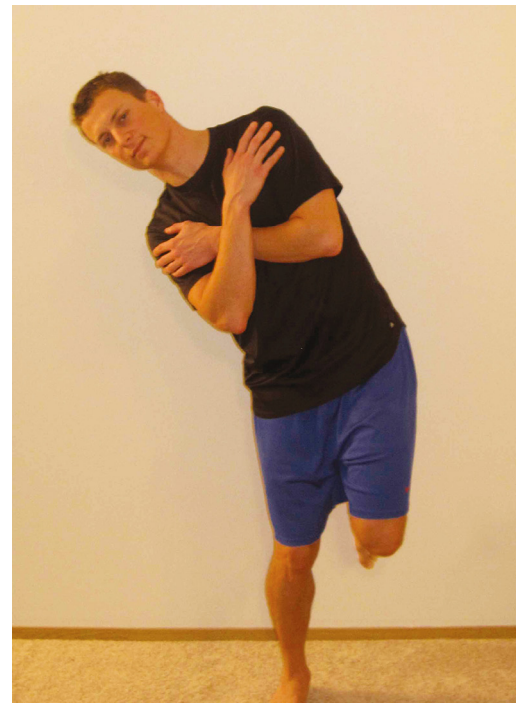
Proprioception and neuromuscular control (NMC) are vital and often unrecognized components of athletic activity, and loss of these is increasingly noted as a predisposing factor for injury. Commonly called muscle memory, these abilities help an athlete to control the limbs throughout the full spectrum of motion. Deficits here can be subtle and difficult to assess in the typical office visit. Testing must be multiplanar and mimic the athlete's typical motion patterns. The examining physician must have a thorough understanding of the motions involved in the athlete's sport to make the appropriate changes; for example, a difference of 10 degrees in knee extension during the cocking phase of a tennis serve increases the valgus load at the elbow by 21%.⁶⁸ When necessary, this can be accomplished through discussion with the athlete, parent, coach, and/or trainer.

Prehabilitation

Prehabilitation is based on the concept that many sports injuries can be prevented if the athlete engages in an appropriate pre-season prehabilitation program. Most of the scientific literature



• **Fig. 40.2** Single leg squat.



• **Fig. 40.4** Frontal plane evaluation.



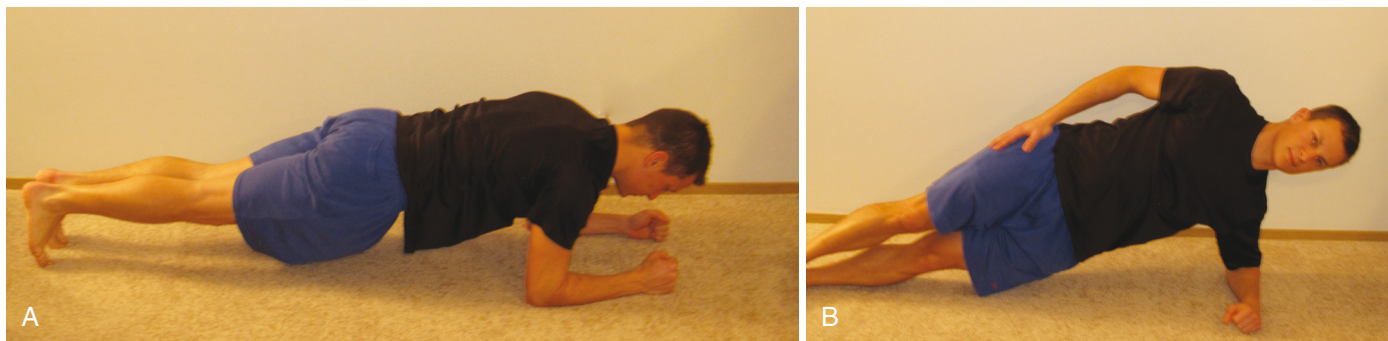
• **Fig. 40.3** Sagittal plane evaluation.

regarding prehabilitation programs focuses on noncontact anterior cruciate ligament (ACL) tears.^{102,103,143} Balance perturbation, plyometric training, and stretching programs all have a role in the prehabilitation of the athlete. Ideally, some of these issues can be assessed during the preseason physical examination and corrected before the start of the season. A large proportion of sports injuries are related to overuse, and it is wise to have a discussion of these

issues with athletes and coaches in the preseason. Off-season cross-training, core muscle work, and cardiovascular preparation are generally considered keys to a healthy season. Prehabilitation can be viewed as an injury risk-reduction program. These programs are sport specific, and often position specific, but they always address the basic components of all athletic movements and the potential “breaks” in the kinetic chain: flexibility, strength, and endurance. These programs can be used during preseason, postseason, or in-season to treat previous injury or to reduce risk of future injuries. For example, a soccer prehabilitation program could involve daily postactivity stretching, strength and plyometric training three times per week, and cardiovascular exercise three times per week. A thrower’s prehabilitation program might involve a focus on daily postexercise ROM, strength training two times per week, and daily technique practice, with pitch/throw volumes increasing weekly. Additional focus can be placed on any component based on the individual athlete’s needs.

Injury Phases

Rehabilitation is guided by the status of the injured tissue, so an understanding of the timing of injury and recovery is essential. Injuries can be acute, subacute, chronic, or an exacerbation of a chronic condition. A detailed history is required because athletes will present at varying points of their tissue healing, and the treatment plan will have to be created appropriately. The physician caters the treatment to the individual tissue and the timing of the injury. The first phase of injury involves the initial injury and the subsequent inflammation, edema, and pain. This phase is typically short, lasting days, depending on the severity of the injury. The reparative phase varies depending on the type of injured tissue but is generally expected to last from 6 to 12 weeks. It involves cell proliferation, granulation tissue formation, and neo-vascularization. The last phase is remodeling, which occurs as the tissue matures and realigns. Excessive inflammatory response or



• Fig. 40.5 (A) Plank. (B) Side plank evaluation.

TABLE 40.2 Typical Return-to-Play Phases

Phase	Goal	Timing
Phase 1 (acute)	Allow injured tissue time to heal, decrease symptoms, maintain range of motion	<72 h
Phase 2 (recovery)	Increase demands on the athlete; flexibility, strength, endurance training; kinetic chain corrections; maintain cardiovascular fitness through cross-training	Variable
Phase 3 (functional)	Advance toward full return to play, advance cardiovascular fitness	2–4 wk

exuberant repair can result in poor outcomes. For example, problems experienced during the remodeling phase can result in excessive scar tissue formation and development of recurrent/chronic injury. Many athletes return before the tissue itself is adequately healed and experience reinjury or an exacerbation of their previous symptoms. This likely occurs if the injured tissue is not fully healed or if the new tissue has not adapted to tolerate the desired motion or inherent stresses of the sport.

Stages of Rehabilitation

Rehabilitation can be viewed as a three-stage spectrum from acute to recovery and then to the final functional stage through return to full play (Table 40.2).¹²¹ Each stage has individual goals that lead to the overall goal of RTP. The acute stage of rehabilitation is focused on managing the symptoms and signs of the injury. Although the classic PRICE (protection, rest, ice, compression, elevation) approach is often followed, a newly widely adopted approach addresses both acute and chronic phases of recovery. The first part of the new approach, PEACE (protection, elevation, avoid antiinflammatories, compression, education), refers to acute treatment. The principles of the acute PEACE phase include a short period (1–3 days) of reduced mobility to reduce bleeding and tissue distention. Although evidence to support elevation is weaker, the recommendation remains to elevate the injured structure above the level of the heart to promote interstitial fluid flow out of the tissue. Avoiding antiinflammatories is a newer addition to the recovery protocol and is based on the principle that inflammation plays a key role in tissue healing and thus inhibiting inflammation, especially

with high-dose antiinflammatory medications, may result in long-term negative consequences to the tissue.⁶⁵ Although there is not definitive evidence to show specific long-term negative outcomes on tissue healing in elite athletes, antiinflammatory medications should always be prescribed at the lowest effective dose for the shortest period of time.⁹⁴ Additionally, cryotherapy, an extremely popular form of analgesia, is no longer recommended because it may lead to a disruption in tissue healing and overproduction of collagen caused by disruption in pathways of inflammation, angiogenesis, neovascularization, and macrophage and neutrophil infiltration. Compression, or application of external mechanical pressure, reduces bleeding and edema. Finally, educating patients on the benefits of active recovery and a realistic timeline for recovery enhances self-efficacy and reduces dependence on passive modalities, which may be counterproductive to long-term healing.⁶⁵

The recovery stage is often the longest part of the rehabilitation program and shifts the focus from symptom management to restoration of strength and endurance as it applies to function. The second part of the new approach, LOVE (load, optimism, vascularization, exercise), serves to guide the physician's treatment approach during the recovery phase. Load refers to optimally loading the affected tissue to promote healing and recovery.⁶⁵ Progressive loading programs are the foundational treatment for various tendinopathies and bone stress injuries. Optimism refers to the psychological aspects of a return to sport. The treating sports medicine team should provide support and encouragement to the healing athlete while steering the athlete away from behaviors that negatively impact a return to sport such as fear-avoidance or catastrophizing. Vascularization refers to an idea of promoting aerobic activity (i.e., vascularizing the healing tissue through aerobic exercise). Cross-training techniques, such as pool aerobics, can be used in place of weight-bearing activities that may be unavailable to the healing athlete in the moment, such as a runner recovering from a foot injury. Lastly, exercise refers to a shift in therapy from passive to active treatments. The kinetic chain is addressed during this time as well, with a focus on flexibility, strength, symmetry, and proprioception. Exercises traditionally advance from closed-chain movements to more open-chain, dynamic movements as the athlete's rehabilitation progresses. Open-chain exercises are ones in which the distal segment is free, whereas closed-chain exercises are ones in which the distal segment is fixed. In the case of an athlete with a shoulder injury, the rehabilitation program moves from predominantly closed-chain exercises, to axial-loaded exercises, to open kinetic chain exercises, to full sport-specific rehabilitation.¹⁵⁵ Kinetic chain rehabilitation is modified to the individual athlete's

symptoms and dysfunction. Rather than view this type of rehabilitation as a series of discrete steps, the physiatrist should consider it as a transition from dysfunction to function and adjust the treatment plan and timing accordingly.¹¹⁰

The parameters of the strength training program (repetitions, resistance, speed, multijoint movements) can be adapted during this time, depending on the clinical scenario. Preexisting kinetic chain issues or tissue overload issues from the current injury are addressed at this time as well. As the rehabilitation program continues, increasingly athletic maneuvers (running, jumping, cutting) are used. NMC involves proprioception, muscle control, and the interplay between them. Without sufficient NMC, the athlete might continue to be predisposed to future injury. This recovery phase is the most variable and has the highest potential for reinjury because the tissue is actively remodeling. The athlete must be reevaluated frequently during this phase because setbacks are common and can require modification of the rehabilitation plan. The athlete can advance to the final stage once pain-free ROM is achieved and strength is at least 80% compared with the noninjured side.

During the final functional stage of rehabilitation, the focus remains on kinetic chain issues and technique errors. Strength, balance, power, endurance, functional ROM, and NMC are aggressively addressed. Sport-specific drills are used during this stage and advanced to include practice. Full RTP is achieved when the injury is no longer painful; when there is normal flexibility, strength, and proprioception; and when appropriate sport-specific mechanics and sport-specific skills are achieved and reproducible. In many cases, this rehabilitation phase becomes the maintenance program.⁸⁰

Biomechanics of Sports

Throwing

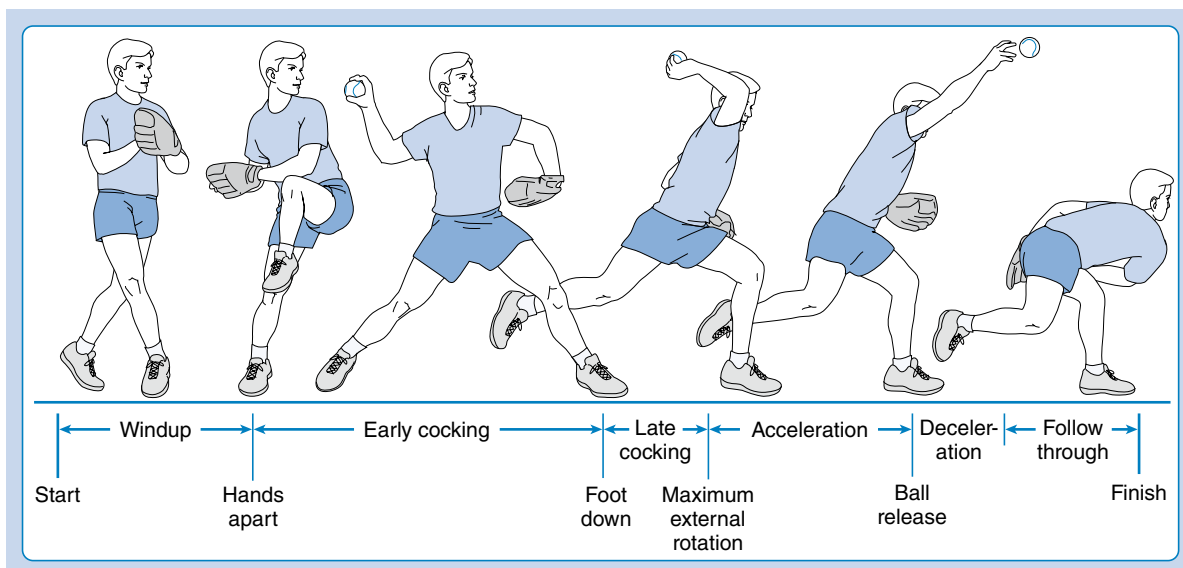
Pitching a baseball is a classic example of the body acting as a kinetic chain of motion, where foot placement and ankle ROM will affect how the pitcher uses the shoulder girdle and subsequently throws the ball. If there is a “weak link” in the pitcher’s kinetic chain, there can be a predisposition to injury because of the repetitive nature of pitching. Whole body activity is

involved in transferring momentum from the body to the ball. The momentum begins with a drive from the large leg muscles and rotation at the hips, progressing through segmental rotation of the trunk and shoulder girdle, transferring energy through elbow extension, next through the small forearm and hand muscles, and finally to the ball. This transfer of momentum and subsequent energy is related to a series of body segment accelerations and decelerations. When an initial segment of the body (e.g., trunk) accelerates, the next segment (e.g., arm) within the kinetic chain is left behind. When the trunk decelerates, the momentum is transferred to the arm with an increased velocity of the arm that is accentuated by the forces acting on the shoulder/arm. This motion and the forces are ultimately transferred to the ball. Approximately 50% of the velocity of a pitch results from the step and body rotation (from the potential energy stored in the large leg and trunk musculature).^{124,224} The other 50% comes from the smaller muscles of the shoulder, elbow, wrist, and hand. When forward stride is not allowed, the peak velocity of a pitched ball decreases to 84%. When the lower body is restricted, it decreases to 64%. Peak velocities in water polo are approximately 50% that of baseball because of the lack of a ground reaction force.

The baseball pitch is composed of six phases (Fig. 40.6). Understanding these phases is critical to understanding, diagnosing, preventing, and treating throwing injuries.

Windup begins when the pitcher initiates motion and ends with maximum knee lift of the lead leg when the ball is removed from the glove. Push-off of the stride leg allows the center of gravity of the body to move forward. Few injuries occur in this early phase because the shoulder musculature is relatively inactive, and most forces arise in the lower half of the body. The transition from windup to early cocking is the “balance point” at which time the pitcher’s center of gravity is stable.

Early cocking is next and is often called the stride phase.¹⁵⁷ The stride leg extends toward the batter, as does the knee and hip of the pivot leg, propelling the body forward into the stride. As the hips rotate forward, the trunk remains stable, storing elastic potential energy, and the throwing shoulder abducts, extends, and externally rotates, leaving the shoulder in a semicoiled position. Again, in this early phase, there is less risk of injury because most



• Fig. 40.6 The six phases of the baseball pitch.

forces are still being generated in the trunk and lower limbs; however, shoulder activity becomes more evident.⁷⁰ The trapezius and serratus anterior demonstrate moderate to high activity as they begin to protract and upwardly rotate the scapula. The middle deltoid generates the abduction force, and the supraspinatus fine-tunes humeral head positioning within the glenoid. This phase ends when the stride foot makes contact with the ground.

The next phase is *late cocking*, when maximal shoulder external rotation is obtained. The trunk now rotates forward toward home plate, leaving the throwing arm behind. The scapula continues to retract and upwardly rotate, allowing the shoulder to externally rotate from 50 degrees to a maximum of 175 degrees. This extreme amount of rotation allows the greatest accelerating force to the ball over the greatest possible distance. The amount of external rotation obtained correlates with the speed of the pitched ball. Most injuries occur in the late cocking and deceleration phases of throwing (Box 40.2).¹⁷⁸ For late cocking, this is caused by the forces needed to stabilize the shoulder in this extreme ROM. The dynamic stabilizers of the anterior shoulder (long head of the biceps, subscapularis, and pectoralis major) are very active in this phase. The static stabilizers (glenohumeral ligaments, capsule, and labrum) are active as well. The glenohumeral ligaments and capsule increase in laxity because of the extreme ROM in the overhead athlete. This laxity is necessary for performance; however, overstretching these ligaments enhances the work of the dynamic stabilizers with a resultant potential for injury.

The *acceleration phase* begins after the point of maximal shoulder external rotation. At this point, powerful elastic forces of the anterior capsule and musculature internally rotate the shoulder from 175 degrees to 90 to 100 degrees at ball release. Shoulder abduction is relatively fixed in all throwers at 90 degrees. Trunk lateral flexion (and not shoulder abduction) generally determines what position the arm is in relative to the vertical plane. “Over the top” throwers have a greater amount of contralateral trunk flexion, and “sidearm” throwers have less. If there is less than 90 degrees of abduction as a result of fatigue, weakness, or just poor form, the common observation is a dropped elbow. Hanging or dropping the elbow results in a reduced pitch velocity and an increased risk of injury to the shoulder (rotator cuff) and the elbow (ulnar collateral ligament injury). Another common injury in the acceleration phase, particularly in older pitchers, is from subacromial impingement as they internally rotate and adduct the abducted arm during acceleration.

The *deceleration phase* is manifested by large eccentric muscular forces of the posterior shoulder girdle to decelerate that rapid internal rotation of the acceleration phase.^{26,70} Deceleration begins after ball release and ends when the arm reaches 0 degrees of internal rotation. The posterior shoulder girdle is active in this phase, including the lower trapezius, serratus anterior, rotator cuff external rotators (particularly the teres minor), and the posterior deltoid. The biceps is also at peak activity to decelerate rapid elbow extension. Given the high tensile load of eccentric contractions, injury is common in this phase (Box 40.3).

The final phase of pitching is the *follow-through phase*. This is a passive phase in which the body is merely catching up with the throwing arm. Low-grade eccentric loading of the shoulder musculature and little risk of injury are noted during this phase.

Because of the high risk of overuse injury in the young pitcher's throwing arm, age-based pitch count guidelines have been developed to decrease this risk (Table 40.3)¹¹⁷ (<https://www.mlb.com/pitch-smart/pitching-guidelines>). For similar reasons, curve balls and sliders should not be pitched until the athlete reaches

• BOX 40.2 Common Injuries in the Late Cocking Phase of Pitching

- Anterior instability
- Internal (posterior-superior) impingement
- Type II SLAP lesion
- Articular surface rotator cuff tears
- Bicipital tendinopathy
- Medial tension injuries at the elbow
- Ulnar collateral ligament injury

• BOX 40.3 Common Injuries in the Deceleration Phase of Pitching

- Posterior instability
- Isolated rotator cuff tears
- SLAP lesions
- Bennett lesion
- Valgus extension overload syndrome

puberty. It is also recommended that young pitchers not compete in baseball more than 8 months per year. During those four “off” months, they should not compete in other overhead arm sports, such as competitive swimming or javelin throwing.

Running

Notable differences are observed between a walking gait cycle and a running gait cycle.¹⁷⁵ One in particular is a third phase in running called the float phase. Float is a period when neither foot is in contact with the ground. It occurs at the beginning of initial swing and the end of terminal swing (Fig. 40.7). The walking gait cycle has a period of double limb support, which is not present in running; it occurs in the first and last 10% of the stance phase. The stance phase of gait is decreased from 60% while walking to 30% to 40% while running and 20% while sprinting. Consequently, as speed increases, the stance phase decreases. As speed increases, the velocity and range of lower limb motion increase, which in turn serve to minimize vertical displacement and improve efficiency. The body lowers its center of gravity by increasing hip flexion, knee flexion, and ankle dorsiflexion; consequently, the major kinematic difference between walking and running occurs in the sagittal plane of motion.

Faster running also changes foot contact.²⁴⁶ In slower running and walking, contact is typically heel to toe. As running speed increases, foot strike occurs with the forefoot and heel simultaneously or when the forefoot strikes initially followed by the heel lowering to the ground. In sprinting, the athlete maintains weight bearing on the forefoot from loading response to toe-off. The angle of gait is different between walking and running. This is the angle between the longitudinal bisection of the foot and the line of progression. This angle is 10 degrees in walking. In running, it approaches 0 degrees as foot strike is on the line of progression to allow more efficient locomotion by limiting deviation of the center of gravity. Barefoot running, simulated by running in minimalist shoes (shoes with decreased heel cushioning and a “zero drop” from heel to toe), gained popularity after multiple publications proposed that humans evolved to run long distances barefoot and that returning to this style may help to prevent injury.^{24,135}

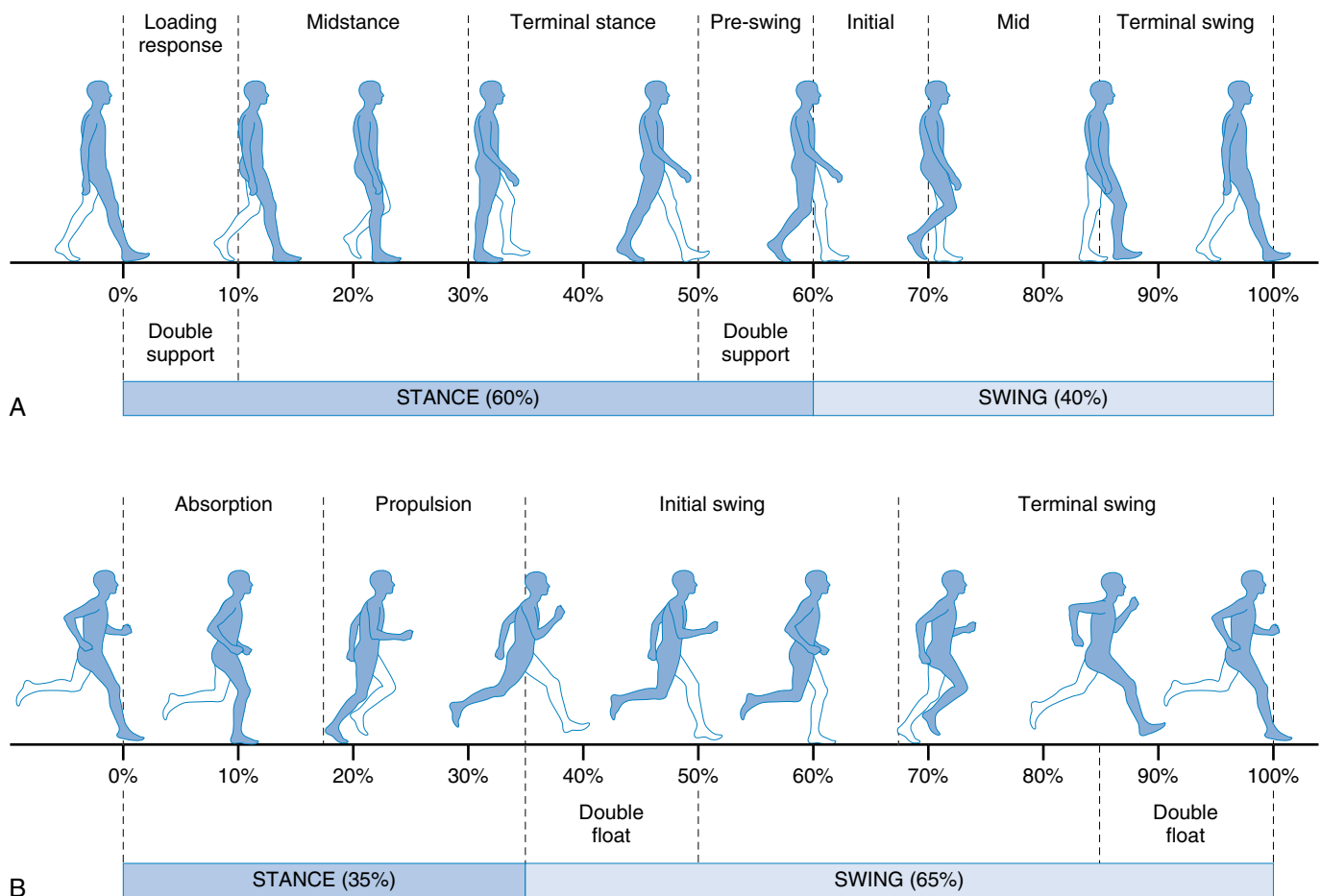
TABLE 40.3 2018 MLB and USA Baseball Age-Based Pitch Counts

Age	Pitch Distance	Max Pitches Per Game	Required Rest by Pitch Count						Max Innings Per Year
			0 days	1 day	2 days	3 days	4 days	5 days	
7–8	46 ft	50	1–20	21–35	36–50	N/A	N/A	N/A	60
9–10	50 ft	75	1–20	21–35	36–50	51–65	66+	N/A	80
11–12	50 ft	85	1–20	21–35	36–50	51–65	66+	N/A	80
13–14	60 ft	95	1–20	21–35	36–50	51–65	66+	N/A	100
15–16	60 ft	95	1–30	31–45	46–60	61–75	76+	N/A	100
17–18	60 ft	105	1–30	31–45	46–60	61–80	81+	N/A	100
19–22	60 ft	120	1–30	31–45	46–60	61–80	81–105	106+	N/A

MLB, Major League Baseball.

If a 13-year-old pitcher who is allowed to throw a maximum of 95 pitches per game throws 66+ pitches, then the pitcher requires 4 days of rest; but if the pitcher throws 51–65 pitches, then the pitcher requires 3 days of rest. The pitcher is allowed to throw up to 20 pitches without any rest required.

Modified from MLB Pitch Smart 2018 Pitching Guidelines. <https://www.mlb.com/pitch-smart/pitching-guidelines>.



• **Fig. 40.7** A comparison of the walking (A) and running (B) gait cycles. (Modified from Ounpuu S. The biomechanics of walking and running. *Clin Sports Med.* 1994;13:843-863.)

Over the past decade, there have been many studies of the biomechanical, clinical, and performance outcomes of barefoot running, but even after multiple systematic reviews with meta-analyses, there remains debate.^{3,105,166,185,253} Proponents argue the following ideas:

- Humans are maladapted to wear (cushioned) shoes, which encourage a rear-foot–strike (RFS) gait by plantarflexing the foot and allowing for a comfortable heel strike.^{76,134}
- Barefoot/minimalist running promotes an evolutionary advantageous fore-foot–strike (FFS) pattern.
- FFS decreases vertical loading rates.^{3,136} Although concomitant increased stride cadence and shortened stride length have been considered as confounders, evidence suggests that FFS is more effective at reducing vertical load rates.⁸²
- Decreased loading rates could potentially lessen overuse injuries in runners, including patellofemoral pain, plantar fasciitis, anterior compartment syndrome, and tibial stress fractures.²²⁰

Others counter with the following ideas:

- Barefoot running is not instinctively acquired by habitually shod runners.
- Many FFS runners transition to RFS when fatigued, and RFS running while barefoot produces the greatest vertical loading rates.
- Barefoot/minimalist running may increase the injury risk of the metatarsals and Achilles tendon.²²⁰
- Barefoot running may increase the risk of tibial stress injury during periods of high training loads.¹¹⁶

Given a lack of high-quality evidence, prescriptions for barefoot running should be cautiously administered and diagnosis specific, with instruction that the patient transition gradually and avoid RFS, especially when fatigued.

Swimming

Four competitive strokes are used in swimming: freestyle (crawl), backstroke, butterfly, and breaststroke. Four phases of the swim stroke are common to freestyle, backstroke, and butterfly. The first phase is the *entry* or *catch*. This encompasses hand entry into the water until the beginning of its backward movement. The *propulsive phase* is divided into two separate phases: pull and push. The *pull phase* ends as the hand arrives in the vertical plane of the shoulder. During the *push phase*, the hand is positioned below the shoulder and pushes through the water until its exit from the water historically at the level of the greater trochanter. The final phase is the *recovery phase*, which entails the aerial return of the hand.

During the propulsive phases of swimming, the arm is moved through adduction and internal rotation starting from a stretched position of abduction and external rotation. The pectoralis major and latissimus dorsi are the major contributors to this motion; however, the serratus anterior and the internal rotator function of the subscapularis and teres provide a major assist.^{148,226}

During the recovery phase, scapular retraction is provided by the rhomboids and middle trapezius, whereas shoulder external rotation is provided by the posterior deltoid, teres minor, and infraspinatus. In preparation for hand entry during midrecovery, the serratus anterior and upper trapezius upwardly rotate the scapula for shoulder stabilization.

Kick patterns are also an important part of swimming mechanics. In the flutter kick for the freestyle stroke, the knees should flex only approximately 30 to 40 degrees, and hip flexion should be minimal. The breaststroke kick is a whip-kick that creates a significant valgus moment at the knee. Because of this increased

knee valgus, medial knee injuries in breaststrokers are common. The pain can be from a medial collateral ligament sprain or a medial plica/synovitis.

General rehabilitation and prehabilitation principles have been established for swimmers with shoulder pain, and all sports medicine clinicians should be familiar with them. The overarching principle is that most shoulder pain in swimmers (impingement and rotator cuff tendinopathy) is caused by dynamic muscle imbalances, weakness, and biomechanical faults, rather than hard anatomical factors. A major tenet of shoulder rehabilitation for the swimmer is scapular stabilization, with a prime focus on endurance training of the serratus anterior and lower trapezius. The serratus anterior is a muscle of particular focus because it has been demonstrated to function at 75% of its maximum test ability in swimming and is active throughout the swim stroke cycle.²⁰⁶ Other tenets of shoulder rehabilitation in swimmers include stretching the internal rotators and posterior capsule and cervical and thoracic mobilization.

Jumping and Landing

The biomechanics of jumping and landing in sports is particularly well researched in the setting of noncontact ACL injuries of the knee. Noncontact ACL injuries occur more frequently with the knee in less flexion.²⁵² In this position, there are greater knee extensor loads, with greater forces creating anterior tibial translation and subsequent ACL injury. A well-defined sex difference observed in jumping and landing mechanics is likely one reason for the higher rate of noncontact ACL injuries in female athletes.³⁸ Female athletes land more erect with less knee and hip flexion and less hip external rotation and abduction. They also generally have an imbalance of increased quadriceps to hamstrings activation ratio, creating greater knee extension and lesser knee flexion forces. One goal of ACL injury prevention programs is to improve jumping and landing technique by increasing knee and hip flexion during landing and balancing the quadriceps to hamstring activation ratio. A metaanalysis of metaanalyses of ACL injury reduction training programs showed an overall 50% reduction in risk of all ACL injuries in all athletes, and 67% reduction in noncontact ACL injuries in females.²³⁸ Prevention is not time consuming because randomized controlled trials (RCTs) have shown a 64% to 75% reduction in ACL injuries with as little as 15-minute warmups performed twice weekly.^{211,231}

Pharmacology in Sports

Doping refers to any substance or method used to increase performance, possibly to the detriment of the health of the athlete or the ethics of the competition.⁵⁵ WADA was created to unify the fight against doping. WADA rulings are accepted by all National Olympic committees, the International Olympic Committee (IOC), International Paralympic Committee (IPC), national governments, and international sports federations. The agency, based in Montreal, Canada, consists of government representatives from all continents and members of the Olympic movement. The World Anti-Doping Code was developed in 2003 and enforced in 2004, with a complete listing of banned substances and methods.⁵⁵ A brief history of doping in sports is presented in [Box 40.4](#). A summarized list of substances banned by WADA is available online.²⁴⁸ Banned substances are divided into those that are banned in and out of competition and those that are banned in

• BOX 40.4 History of Doping and Drug Monitoring

- World War II: Steroids used in the belief that male hormones could benefit the emaciated prisoners rescued from concentration camps
- 1950–1960s: Amphetamines and anabolic steroids widely used by athletes
- 1962: International Olympic Committee passed a resolution against doping, banned steroid use
- 1967: Medical commission started medical controls in 1968 Olympics
- 1976: Anabolic steroids controlled for the first time in Olympics
- 1988: First gold medal stripped in track and field because of doping
- 1990: Anabolic Steroid Act passed
- 1998: World Anti-Doping Agency formed in Montreal, Quebec, Canada
- 2003: World Anti-Doping Code is passed

competition only.²⁴⁹ The lists of drugs and supplements banned by the National Collegiate Athletic Association (NCAA) and other organizations are available online.¹⁶⁷

When an athlete has an illness or condition that requires treatment with a banned medication, a therapeutic use exemption (TUE) can provide the authorization to take the needed medicine. Per WADA, all international federations and national antidoping organizations must have a process in place whereby athletes with documented medical conditions can request a TUE.⁷⁷

Following is a brief overview of therapeutic and performance-enhancing drugs and supplements, all of which are given careful consideration by these agencies.

Therapeutic Drugs

Analgesics

Acetaminophen may be linked to decreased muscle building after exercise (to an extent similar to that of ibuprofen). Prostaglandins are normally released after eccentric resistance exercise. This response may be blunted after consumption of maximal doses of ibuprofen or acetaminophen, potentially influencing the anabolic response of muscle to this type of exercise.²²⁵

Antiinflammatories

One study found that 15% of high school football players used nonsteroidal antiinflammatory drugs (NSAIDs) daily.²³⁴ Although the pain relief provided by NSAIDs may enhance performance, NSAIDs are not considered ergogenic. NSAIDs can mask pain and interrupt a natural defense mechanism for preventing injury. Because of their antiinflammatory effects, NSAIDs likely inhibit the production of prostaglandin E₂, which is known to play a role in bone healing.¹⁸⁷ NSAIDs have been linked to nonunion after fracture when taken for longer than 4 weeks.² Consequently, in those with suspected or known fractures, NSAIDs should be used cautiously.

Corticosteroids are potent antiinflammatories. In general, when given orally, rectally, intravenously, or intramuscularly, their use is prohibited by WADA. Topical preparations for skin, eye, ear, nose, or buccal cavity or for iontophoresis are allowed. For applicable athletes, epidural and intraarticular steroid injections, as well as inhaled steroids, require a TUE.²⁴⁹

Antihypertensives

Diuretics are often a first-line treatment of hypertension in the general population.²⁴² Angiotensin-converting enzyme inhibitors or calcium channel blockers can be considered first-line agents for

treatment of hypertension as well, particularly in athletes, because diuretics and beta blockers have side effects that may hinder performance. The detection and management of hypertension and the efficacy of various agents are beyond the scope of this chapter. Rather, this discussion will focus on the aspects of these medications that affect athletes.

Diuretics can decrease plasma volume, cardiac output, and peripheral vascular resistance. Dehydration and electrolyte alterations may result in cramps or even heat stroke. For amateur athletes, the benefits of antihypertensive therapy with diuretics likely outweigh the risks. In the case of elite athletes, diuretics are banned in part for their theoretic ability to increase urine output and mask use of other banned agents (although this method rarely works optimally).¹⁸⁷

Beta blockers can reduce exercise tolerance by increasing perceived effort. They can also inhibit glycolysis and glycogenolysis with resulting hypoglycemia after exercise. The negative chronotropic effects of beta blockers can decrease heart rate recovery after exercise. Beta blockers are banned in certain sports during competition because of their anxiolytic effects.¹⁸⁷

Diabetes Drugs

Insulin doses may need to be adjusted for persons with insulin-dependent diabetes starting a new exercise program. A 20% to 40% reduction is typical because of increased insulin sensitivity with exercise. High-intensity exercise (i.e., >80% VO₂ max) can cause a temporary increase in blood glucose secondary to increased sympathoadrenal activation. In that situation, supplemental insulin, if used, should be given at a smaller dose than that given for hyperglycemia at rest.^{16,112}

Intramuscular injections of insulin should be avoided because muscle contraction can accelerate insulin absorption. Heat can increase absorption rates of insulin, although cold can decrease them. Therefore athletes with insulin-dependent diabetes mellitus should avoid modalities that use extremes of temperature, such as hot or cold whirlpools. Extreme ambient temperature can also reduce insulin action in athletes.¹¹² Insulin is banned because it is a hormone and metabolic modulator, and a TUE is required for all insulin-dependent individuals with diabetes.²⁴⁸

Asthma Drugs

Exercise-induced bronchospasm (EIB) can be treated effectively with a short-acting beta agonist (SABA), such as albuterol, 15 to 20 minutes before exercise.^{216,240} If not sufficient to prevent symptoms, cromolyn (a mast cell stabilizer) can be added. Inhalation of either cromolyn sodium or a beta agonist, or both, 15 minutes before exercise is almost always successful in blocking EIB and airway inflammation.¹³⁰ A TUE can be granted for use of a beta agonist if an asthma diagnosis is established.²⁴⁸

For patients who have chronic persistent asthma (forced expiratory volume in 1 second <80% of predicted and symptoms greater than twice a week), inhaled corticosteroids (ICSs) are standard treatment. ICSs do not appear to have ergogenic or anabolic effects.¹⁸⁷

Performance-Enhancing Drugs

Anabolic Steroids

It is estimated that 3 to 4 million athletes in the United States alone have used anabolic steroids (ASs),⁵ with annual market sales well in excess of \$100 million.²²³ In 2022, an estimated 0.8% and 1.3% of 8th and 12th graders, respectively, reported misuse of

steroids in the previous 12 months.¹⁶⁹ A study from 2007 found that 6.3% of high school varsity football players were current or former AS users.⁸⁷ ASs have three general effects that enhance athletic performance: (1) By binding androgen receptors, ASs stimulate messenger ribonucleic acid synthesis, increasing structural and contractile protein synthesis^{131,223} and producing an anabolic state.³⁰ (2) ASs are also anticatabolic via competitive inhibition of the glucocorticoid receptor, inhibiting the catabolic effects of cortisol²²³ and preserving muscle mass.³⁰ (3) ASs have emotional effects, pushing athletes to train more intensely and more often.³⁰

Because ASs are illegal, studying their health risks and side effects is a challenge. Inconsistent formulations, dose administrations, and training habits make it difficult to derive statistically sound conclusions. Some studies show minimal effects on body composition and strength, although others show that supraphysiological doses of testosterone (or its derivatives) can lead to an increase in lean mass and muscle size in humans (Table 40.4).²²³

TABLE 40.4 Negative Effects of Anabolic Steroids

System	Effects
Cardiovascular	Increases in total LDL Decreases in HDL Hypertension Myocardial ischemia (imbalance between myocardial oxygen supply and demand) Myocardial infarction Cerebrovascular accident
Hepatic	Hepatocellular dysfunction (abnormal liver function tests) Peliosis hepatis (benign intrahepatic vascular disorder) Hepatocellular carcinoma
Male reproductive	Oligospermia, azoospermia Gynecomastia Decreased testicular size
Female reproductive	Decreased LH, FSH Decreased progesterone Menstrual irregularities Male pattern alopecia Hirsutism (irreversible) Clitoromegaly (irreversible) Deepening voice (irreversible)
Skeletal	Premature closure of the epiphyses
Skin	Acne (increased number of sebaceous glands) Striae
Psychological	Decreased libido Mood swings Hypermania/hypomania Aggressive behavior (“roid rage”) Withdrawal Depression Addiction

FSH, Follicle-stimulating hormone; HDL, high-density lipoprotein; LDL, low-density lipoprotein; LH, luteinizing hormone.
From Laos C, Metz J. Performance-enhancing drug use in young athletes. *Adolesc Med Clin.* 2006;17:719-731.

Typical users of steroids can take 10 to 40 times the prescribed dosage for disease states.^{87,110,149,223} Premature deaths have resulted from AS use, most commonly as a result of suicide and acute myocardial infarction.¹¹⁰ The Anabolic Steroid Control Act of 1990 prohibited the use of steroids for any use other than disease treatment, thereby classifying ASs as schedule III drugs within the Controlled Substances Act. ASs are banned by all major sporting leagues, the IOC, and the NCAA.^{149,167,249}

Erythropoietin and Blood Doping

Endurance athletes are particularly sensitive to the oxygen-carrying capacity in their blood. For years, athletes sought a competitive advantage by training at altitude or sleeping in altitude tents. This effect can be reproduced by blood doping, blood transfusion, or administration of the drug recombinant human erythropoietin (rhEPO).^{35,223} EPO is a glycoprotein hormone produced mainly in the kidney in response to tissue hypoxia. EPO is part of a negative feedback cycle that controls tissue oxygen delivery by controlling the number of erythrocytes in the blood.¹⁹⁸ Both rhEPO and transfusions have been shown to increase VO₂ max.^{19,223} The risks of artificially elevated hemoglobin/hematocrit include stroke, myocardial infarction, and pulmonary embolism. Blood transfusions, even conducted according to hospital procedures, carry risks for infection, including HIV and hepatitis, as well as transfusion reaction.²²³

Stimulants

Stimulants might be the most common and underrecognized supplement used by high school athletes. Common stimulants include caffeine, ephedrine, pseudoephedrine, phenylephrine, amphetamines, and methamphetamines.⁸⁷ Stimulants increase arousal, respiratory rate, heart rate, and blood pressure. Side effects include dizziness, insomnia, agitation, restlessness, anxiety, confusion, paranoia, hallucinations, dyskinesias, gastrointestinal disturbances, heat intolerance, stroke, myocardial infarction, arrhythmia, and death.⁸⁷

Pseudoephedrine (Sudafed) is a commonly used drug in this class. Although improved cycling power output¹⁴ and pace in a 10-km run¹⁵ have been demonstrated in studies of this drug, other trials contradict these findings.⁸⁴ Pseudoephedrine is a chemical precursor in the illicit manufacture of methamphetamine. Federal policies currently restrict sales by limiting purchase quantities to consumers of a minimum age.⁷⁵ Pseudoephedrine is subject to a threshold limit and is banned by WADA when its concentration in urine is greater than 150 µg/mL.²⁴⁹

The herbals ephedra and ma huang have similar structural properties and physiological effects to pseudoephedrine.⁷⁵ Ephedra and ephedra-containing products were taken off the market in 2004.¹⁴⁹ Ephedra is banned by the NCAA, IOC, and major league organizations.^{149,167,249}

Caffeine is an adenosine receptor antagonist with stimulant properties.⁷⁵ Studies have shown that caffeine is ergogenic in most, if not all, exercise situations, with few negative effects during exercise.^{86,88} Caffeine acts by binding to adenosine receptors in most tissues, including brain, heart, smooth muscle, fat, and skeletal muscle, with a wide spectrum of interacting responses. Caffeine also stimulates the secretion of epinephrine⁸⁶ and the central nervous system and enhances peripheral neuromuscular transmission and muscle contractility.⁷⁵ Side effects of caffeine include nausea, heart palpitations, headache, and muscle tension. Caffeine might also act as a diuretic, although no studies have shown a risk for dehydration in athletes.⁷⁵ Caffeine is monitored in competition but has not been prohibited by WADA since 2004.²⁴⁹

Supplements

A drug is any substance that exerts an effect on a body system, but a supplement is defined as a substance taken to augment the diet. Regulatory bodies within the US government handle drugs and supplements differently, and supplement manufacturers are not held to the same standards as drug companies. Supplement manufacturing is overseen by the US Food and Drug Administration (FDA) and treated as food as long as no drug claims are made.⁸⁷ Most experts agree that a well-balanced, isocaloric diet of commonly available food is sufficient to guarantee basic nutritional requirements for most athletes, including macronutrients (carbohydrates, proteins, and lipids) and micronutrients (vitamins, minerals, and trace elements).⁵⁸ Despite this, many athletes take extreme doses of essential nutrients as dietary supplements (amino acids, vitamins, minerals), at times greater than tolerable levels.⁵⁸ The global supplement industry has an estimated value of \$152 billion with an expected value of \$300 billion by 2028.⁶¹ According to the CDC, 57.6% of adults over age 20 years had used dietary supplements within the past 30 days.³⁶ Despite a large number of available supplements on the market, few actually show a sports performance benefit.

Creatine

Creatine (Cr) use is common among athletes with prevalence rates of 9% to 46% reported in adults and similar rates in high

school athletes.⁹⁵ Cr is a naturally occurring compound made from amino acids glycine, arginine, and methionine.^{131,223} It is proposed that Cr phosphate supplementation benefits short-duration, high-intensity, repetitive exercise by enhancing adenosine triphosphate regeneration.^{95,131,187} Common side effects of Cr include muscle cramping and gastrointestinal distress.¹⁸⁹ A number of cases of renal failure have been reversed by withdrawal of Cr supplementation.^{131,223} It is notable that most studies in the literature were short term with healthy subjects. In addition, there are insufficient data on the possible effects of Cr on other Cr-containing tissues, such as the brain, cardiac muscle, or testes.²²³ Because it is not classified as a drug, Cr is not under direct regulation by the FDA and is widely available over the counter.²²³ The NCAA does not allow member teams to provide Cr to its players. For a summary of drugs and supplements used in sports, see [Table 40.5](#).

Preparticipation Physical Evaluation

The primary goals of the PPE are to (1) determine general physical and psychological health of the athlete, (2) evaluate for life-threatening or disabling conditions, including risk of sudden cardiac arrest, and (3) serve as an entry point into the health care system for those athletes who do not have a primary care physician. Added opportunities are to discuss preventative health and high-risk behaviors and to establish rapport with the athlete.

TABLE 40.5 Summary of Drugs and Supplements Used in Sports

Drug/Supplement (Street Name in Parenthesis)	Postulated Effect	Evidence of Effectiveness	Adverse Effects	Legal Status
Anabolic steroids (testosterone or “T,” designer steroids)	Increase lean muscle mass	Effective	Significant, multiple systems	Illegal unless prescribed for disease state
Androstenedione and DHEA (Andro)	Increase testosterone and lean muscle mass	No effect with androstenedione; DHEA with some benefits in patients >70 years old	Possible cardiovascular effects	Prohibited by WADA, NFL, and NCAA; androstenedione is illegal
Beta hydroxy-beta methylbutyrate (HMB)	Decreases protein breakdown; increases synthesis	Mixed	None, but long-term data lacking	Legal
Blood transfusion or erythropoietin (doping, EPO)	Increases endurance and oxygen delivery	Effective	Significant cardiovascular risks	Prohibited by WADA, NFL, and NCAA
Caffeine	Increases energy, systemic effects not fully known	Effective	Minimal	Legal
Creatine	Increases muscle mass and strength	Effective	Few case reports of renal failure; proposed risk of dehydration	Legal
Ephedrine, pseudoephedrine (ma huang, guarana)	Increases energy, metabolism, decreases fatigue	Mixed	Significant cardiovascular, psychiatric risks, and others	Prohibited by WADA, NFL, and NCAA
Human growth hormone	Increase muscle protein synthesis and strength	Lack of evidence	Significant	Illegal unless prescribed for disease state

DHEA, Dehydroepiandrosterone; NCAA, National Collegiate Athletic Association; NFL, National Football League; WADA, World Anti-Doping Agency.

Modified from Bemben MG, Lamont HS. Creatine supplementation and exercise performance: recent findings. *Sports Med.* 2005;35:107-125; Fields J, Turner JL. Performance-enhancing sports supplements. In: *Sports Medicine and Rehabilitation: A Sport Specific Approach*. 2nd ed. Wolters Kluwer Lippincott Williams & Wilkins; 2009; Jenkinson DM, Harbert AJ. Supplements and sports. *Am Fam Physician.* 2008;78:1039-1046.

The PPE is a consensus publication by the major sports medicine associations and is an excellent guide for the team physician.¹⁴¹ Roughly 5% of athletes will be excluded or require further evaluation after the PPE.¹⁶⁰ The PPE should be performed with enough lead time to allow appropriate subsequent investigation before the season begins.

The most common settings for the PPE include individual examinations performed in the physician's office or via station-based examinations. Both office- and station-based assessments exclude or require further evaluation of roughly the same percentage of athletes; however, higher rates of exclusion for musculoskeletal or cardiopulmonary reasons may occur in the office-based setting.¹⁶⁰ Individual examinations have the advantages of privacy and more opportunity to discuss other general health and safety issues.

The medical history portion of the PPE is able to detect 88% of general medical conditions and 67% to 75% of musculoskeletal conditions that affect athletes. The history portion of the PPE includes personal and family history of cardiovascular illness, spinal injuries, concussions, neurological disorders, diabetes, asthma, hematological disorders, and musculoskeletal injuries. Additional screening items are included that address mental health, substance use, and high-risk behavior. The physical exam portion of the PPE should include a structured exam with vital signs and visual acuity testing as well as cardiac examination that evaluates for pathological murmurs as well as stigmata of Marfan syndrome. Box 40.5 provides the recommended history and physical examination.¹⁴¹

Many subspecialties might be involved in the PPE. The lead physician must be determined before the decision-making process begins. It is also vital for individual providers to be able to identify suspicious findings outside their own area of specialty and refer to the appropriate specialists when necessary. A provider who is

unsure of a finding should always refer to a specialist for further investigation or consultation on final clearance to play rather than make the decision independently.

Clearance to Play

The lead physician must make the determination to clear the athlete without restrictions or with recommendations for further evaluation or treatment. Alternatively, athletes may not be cleared for participation in certain sports or may not be cleared for participation in any sport.

Cardiovascular Screening

Sudden cardiac death (SCD) is a rare and devastating condition seen at all levels of sport. The American Heart Association (AHA) recommends that cardiovascular screening include specific questions regarding family history, personal history, and a thorough physical examination (see Box 40.5).¹⁴⁷ Controversy exists in sports medicine regarding screening electrocardiograms (ECGs) and echocardiography. The AHA currently does not recommend a screening 12-lead ECG, echocardiograms, or exercise stress testing for athletes without risk factors.^{141,145,147} The IOC and the European Society of Cardiology recommend a screening ECG as part of the PPE, in part because of different etiologies of SCD between the United States and Europe.⁴⁶ The AHA decision is based on the low prevalence of SCD, limited resources, impracticality of mass screening, and lack of a physician base to interpret the ECG, as well as the consequences of false-positive test results.^{46,145,147} Despite this, numerous nonuniversal programs have been implemented throughout the United States to screen young athletes for cardiac disease. These programs are not opposed by the AHA provided all parties are privy to the risks and benefits of screening.¹⁴⁵ Hypertrophic cardiomyopathy (HCM) is the most common cause of SCD in athletes in the United States. It is an autosomal dominant disease, and detailed family history may detect premature cardiac death and lead to further diagnostic testing in the athlete. Echocardiography is the most effective way of diagnosing HCM, although it has been suggested that approximately 200,000 individuals would need to be screened to identify one with the condition.⁶⁹ However, a study in the *New England Journal of Medicine* showed that in screening 11,168 adolescent soccer players with a health questionnaire, physical examination, ECG, and echocardiography, 42 were found to have cardiac disorders associated with SCD.¹⁴² Five of those 42 were found to have HCM; all five had an abnormal ECG and echocardiogram, but none had an abnormal history or physical. Proponents of universal ECG screening point to the fact that screening with history and physical alone is insufficient to identify cardiac disease, as was the case in the previously mentioned study. The presence of HCM warrants complete disqualification from competitive sports. In the aforementioned study, two of the five athletes returned to sport against medical advice and subsequently died during exercise.

Medicolegal Aspects of the Preparticipation Evaluation

The majority of the responsibilities of a volunteer team physician are covered by Good Samaritan laws. The PPE is a notable exception. A physician may be held liable for malpractice if an athlete is cleared to play despite the presence of a medically contraindicated

• BOX 40.5 American Heart Association 14-Point Cardiovascular Screening of Competitive Athletes

Medical History

Personal

- Chest pain, discomfort, tightness, or pressure related to exertion
- Unexplained syncope or near syncope
- Excessive and unexplained dyspnea/fatigue or palpitations associated with exercise
- Prior recognition of a heart murmur
- Elevated systemic blood pressure
- Prior restriction from participation in sports
- Prior testing for the heart, ordered by a physician

Family

- Premature death (sudden and unexpected, or otherwise) <50 years of age attributable to heart disease in one or more relatives
- Disability from heart disease in close relative age <50 years
- Hypertrophic or dilated cardiomyopathy, long QT syndrome, or other ion channelopathies, Marfan syndrome, or clinically significant arrhythmias; specific knowledge of genetic cardiac conditions in family members

Physical Examination

- Heart murmur
- Femoral pulses to exclude coarctation of the aorta
- Physical stigmata of Marfan syndrome
- Brachial artery blood pressure (sitting position)

condition. Liability waivers and similar documentation are not acceptable in the presence of medically contraindicated conditions. Athletes can legally challenge physician nonclearance decisions and have occasionally won the right to participate despite medical judgment. In the presence of a life-threatening condition, physicians are advised to follow their medical judgment and not clear the athlete for play.

Emergency Assessment and Care

The first tenet of care for a sports medical emergency is preparing, developing, and rehearsing an EAP. The EAP is designed to develop comfort and control in the potential chaos of an emergency. The plan must be reviewed and practiced at least annually with all members of the sports medicine team. Those designated as “first responders” should be adequately trained in cardiopulmonary resuscitation (CPR) and AED use. One goal is to have access to early defibrillation, defined as the first shock given within 3 minutes of collapse. During the initial evaluation and triage, rapid recognition to determine whether the EAP or routine management should be enacted is critical. Having the sports medicine physician on the sidelines of an event is crucial to be able to witness the mechanism of injury. Being an eyewitness allows one to determine whether there was contact or collision involved in the collapse and whether the collapse occurred during activity or after the play was over. Quickly determining whether additional providers or transport is necessary through emergency medical services is important at this stage as well. This section presents the basics of emergency assessment and care, including a review of CPR guidelines and SCD in athletes. It then discusses specific emergency scenarios, including the initial assessment of the athlete with a cervical spine injury in contact sports and EAC in athletic endurance events.

The primary goal of CPR is to maximize blood flow by provision of effective and uninterrupted chest compressions for longer periods. Clinical studies have demonstrated improved survival when 90 to 120 seconds of CPR are used before shock delivery when the rescuer's arrival to the patient is greater than 5 minutes.²⁴³ Perfusing the myocardium with oxygenated blood before defibrillation improves survival. If the collapse is witnessed (and the victim is down <5 minutes), immediate defibrillation is still indicated. The emphasis of CPR has moved away from ventilation and placed further on chest compressions. In some locations, ventilations have been completely dismissed in favor of chest compressions only, begging a name change from CPR to CCR (cardiocerebral resuscitation).⁷¹ CPR guidelines are reviewed and updated every 5 years; therefore it is important for the sports medicine physician to stay abreast of current guidelines.

Sudden Cardiac Death in Athletes

The leading cause of death in young athletes is SCD. In a 20-year review of National Collegiate Athletic Association athletes, the overall incidence of SCD was 1 in 63,682 athlete-years.¹⁸⁶ The highest incidence of SCD was in Division 1 male basketball players (1 in 8188 athlete-years). The most common postmortem finding was autopsy-negative unexplained death at 19.5%. Idiopathic left ventricular hypertrophy or possible cardiomyopathy was next most common at 16.9%. HCM represented 12.7% of cases. Eight cases over the 20-year study period were attributable to myocarditis, none of which were associated with COVID-19 infection. Most athletes with sudden cardiac arrest are found in

asystole or pulseless electrical activity (PEA), both nonshockable rhythms. The next most common are ventricular fibrillation (VF) and ventricular tachycardia (VT) (both shockable rhythms). It is likely that most victims start in VF or rapid VT at the time of collapse, but the rhythm has already deteriorated to asystole or PEA before the first rhythm analysis. The probability of successful defibrillation from VF decreases rapidly over time, with survival rates declining 7% to 10% for every minute defibrillation is delayed, which emphasizes the importance of AEDs and a well-rehearsed EAP.

Survival after SCA is highly dependent on time to treatment and availability of onsite trained personnel. One study showed that with a certified athletic trainer on site, survival rate after SCA was as high as 83%.⁶⁴ Several factors may delay treatment in SCA, and all medical personnel need to understand them. The first is agonal or occasional gasping mistaken for breathing. Second, many lay rescuers falsely identify the presence of a pulse; consequently, current CPR guidelines eliminate lay rescuer assessment of pulse and recommend assuming cardiac arrest if the victim does not demonstrate normal breathing. The third factor is myoclonic activity falsely identified as seizure and not cardiac arrest. Seizure activity is present in approximately 20% of patients with cardiogenic collapse.⁶³

Cervical Spine

Cervical spinal cord injuries (SCI) are one of the most catastrophic injuries in sports medicine. They have the potential to occur in any sport but are predominantly seen in football and ice hockey. A catastrophic cervical SCI is defined as “a structural distortion of the cervical spinal column associated with actual or potential damage to the spinal cord.”¹⁶⁰ There are approximately 250 new sport-related SCI each year, and 85% of these result in tetraplegia. Advances in helmet technology in the 1960s decreased the incidence of intracranial hemorrhage but paradoxically increased the incidence of SCI. Rule changes, including the banning of spear tackling, have been effective at mitigating this rise. The risk of injury can never be fully avoided in contact sports, but the minutes that follow the injury itself are vital to prevent secondary neurological or cardiopulmonary sequelae. Performing evidence-based prehospital care is essential to achieve the best possible postinjury outcomes.

Any athlete who is unconscious or unable to move must be treated as having a SCI or unstable fracture until proven otherwise. As with all injuries, the athlete should be assessed for the need of basic life support. The conscious athlete should be evaluated for neck pain with and without motion as well as for extremity symptoms. Recommendations from a 2021 consensus statement on prehospital management of SCI in athletes include the following: (1) In athletes with suspected SCI, airway access should be established before transport; (2) American tackle football facemasks should be removed before transport to ensure airway access in athletes with suspected SCI; and (3) appropriate tools and trained personnel should be available for facemask removal.¹⁶⁰ In athletes with SCI, maintaining cervical alignment is the highest priority, and studies have shown that removal of the helmet alone in American football, ice hockey, and lacrosse significantly affects cervical spine alignment, whereas removal of both the helmet and shoulder pads results in smaller but statistically significant alignment changes. Recommendations for helmet and shoulder pad removal is left to the expertise of the personnel on the field, but the removal must be done by those trained in the

specific equipment.¹⁶⁰ Transferring the patient is another important consideration for an athlete with a suspected SCI. Again, the trained professionals at the scene should determine the optimal transfer technique. When feasible, the eight-person lift and slide technique is preferred for supine athletes, while the log-roll push technique is preferred for prone athletes. These transfers are stressful and nonintuitive and should be practiced by team physicians and training staff to be performed correctly.

Exercise-Associated Collapse in Athletic Endurance Events

EAC in the endurance athlete has a broad differential diagnosis (Box 40.6). Given space constraints, this section focuses on benign EAC (postural hypotension), exercise-associated hyponatremia (EAH), cardiogenic collapse, and heat-related illnesses. In general, when an athlete collapses before the finish line in a race while still running, the diagnosis is more ominous. The assessment of a collapsed athlete starts with assessing the level of responsiveness and checking the ABCs: airway, breathing, and circulation. Depending on the severity of the presentation, specific symptoms, and vital signs, a diagnostic workup can include an assessment of cardiac rhythm, rectal temperature, and blood glucose and sodium levels. The most common cause of collapse in a marathon runner after crossing the finish line is benign EAC. This is generally considered a form of postural hypotension. While running, the leg muscles act as a venous pump to improve blood flow return to the central circulation. When the athlete stops running, venous pooling of blood in the legs can occur, resulting in a decrease in blood pressure and a subsequent collapse. This is exacerbated in warm environments because blood flow is shunted from the core to the skin to facilitate cooling. It is important to keep the athlete walking after crossing the finish line to keep the muscular venous pump engaged. Treating EAC means treating the primary cause of collapse and providing supportive care. If the underlying cause of the collapse is determined to be benign EAC, oral rehydration and lying the athlete down on a stretcher with legs and pelvis elevated above heart level are typically all that is needed. If the athlete does not improve within 15 to 30 minutes, a search for a more ominous cause of collapse should be undertaken, including orthostatics, if possible, and an electrolyte assessment, followed potentially by administration of intravenous fluids. It is rare to definitively need intravenous fluids after running a marathon.

EAH was first reported in the 1981 Comrades Run (90-km ultramarathon) in South Africa; it received national and international media attention after the death of a 28-year-old female runner in the 2002 Boston Marathon. It is a hypervolemic hyponatremia causing early symptoms of lightheadedness, a nause-

ated feeling, later a progressive headache, vomiting, confusion, and finally obtundation, seizures, and death. The pathophysiology is fluid shifts from low osmotic pressure in the blood causing cerebral edema and then neurogenic pulmonary edema. Further research into this condition demonstrates similarities to the syndrome of inappropriate antidiuretic hormone, which has an impact in understanding the pathophysiology underlying EAH. The early symptoms of EAH are nonspecific, so medical providers must have an index of suspicion in any marathoner who is not feeling well after the race. Risk factors for EAH include weight gain during the race, marathon race time more than 4 hours, and body mass index extremes.⁴ Those at risk are runners who ingest too many fluids on the racecourse and subsequently gain weight. Slower runners have more of an opportunity to drink excessively, and smaller runners generally need less fluid to dilute their serum sodium levels. Once the diagnosis is made, treatment is indicated according to the severity of presentation and serum sodium level. For those runners who are fluid overloaded with low serum sodium but have minimal symptoms, close observation and fluid restriction are indicated as a natural diuresis occurs. If the sodium level is low or is not reversing, hospital observation might be indicated. If the same athlete has progressive encephalopathic symptoms, treatment would include high-flow oxygen, a bolus of hypertonic saline (3% NaCl), and quick transport to an emergency facility.

Prevention of EAH should be the ultimate goal for any medical director of an endurance athletic event such as a marathon. Prevention starts with educating athletes about the risk of overdrinking on the course and teaching them to use their training in the months before the event to determine their individual fluid needs. Emphasizing individual differences and reminding the athletes to replace what they need (sweat losses) and not necessarily more are appropriate (Box 40.7). The mantra “drink when you are thirsty” is generally safe for slower and at-risk runners. Only if the athlete is a very salty sweater or if the competition lasts more than 6 hours should sodium or electrolyte replacement be necessary. Spacing the water/aid stations approximately every 1.5 miles along the course is another way to limit excessive drinking and potentially prevent EAH.

Given the continued popularity of marathons and other mass participation endurance events, deaths are commonly reported in the media. With more events and more press coverage, there is an appearance of an increase in marathon-related deaths. The often-quoted risk of SCA in a marathon is 1:50,000. This is an old statistic from a study evaluating cardiac-related deaths in the Twin Cities Marathon and Marine Corp Marathon from 1976 to 1994.¹⁴⁶ When follow-up data from these marathons from 1995 to 2004 were analyzed, the new rate of cardiac-related deaths was 1:220,000.¹⁹⁷ Evaluating more than 10 million runners from 2000 to 2010, the Race Associated Cardiac Arrest Event Registry demonstrated a cardiac arrest incidence during long-distance races (marathons and half-marathons) of 1 per 185,000, with 71% of arrests resulting in death (1:259,000).¹²⁶ Higher incidence was

• BOX 40.6 Differential Diagnosis of Exercise-Associated Collapse

- Benign exercise-associated collapse
- Cardiac arrest
- Heat-related illness
- Heat stroke
- Hypoglycemia
- Hyponatremia
- Hypothermia
- Muscle cramps
- Other medical/neurological conditions

• BOX 40.7 General Fluid Replacement Guidelines in a Marathon

- Drink when you are thirsty
- 400–800 mL (14–27 oz) per hour of racing
- The low end is for smaller, slower runners
- The high end is for larger, faster runners

suggested in longer races with arrests occurring more frequently in marathons (1:100,000) than half-marathons (1:370,000). Earlier recognition, better preparation, and AED use have likely resulted in a decreased rate of SCD during the past several decades.

Heat-related illness is another etiology of EAC. Heat exhaustion is the inability to continue to exercise in the heat but is not related to body temperature. It represents the failure of the cardiovascular response to workload, high environmental temperatures, and dehydration. No chronic or harmful effects are known. In contrast, heat stroke is a medical emergency. Heat stroke is defined as multiorgan system failure secondary to hyperthermia. The rectal core temperature in an athlete with heat stroke is generally more than 102.2°F (39°C). Treatment is immediate total body cooling. Various cooling methods exist. Ice bath/cold water immersion is a commonly used and effective cooling strategy. Other cooling strategies include the “taco method,” using wet sheets and ice, or administering ice to the head, neck, axilla, and groin. Whatever method is used, it should be simple and safe, provide adequate cooling, and not restrict other forms of treatment, including CPR, defibrillation, and intravenous cannulation.²⁰ The mortality rate and extent of organ damage in athletes with heat stroke are proportional to the length of time between core temperature elevation and onset of cooling therapy. Finally, it is important to note that heat stroke can occur in cool environments and might be more of a genetic predisposition to excessive endothermy (i.e., endogenous heat production) and not just a factor of extreme environmental conditions.¹⁹⁴

Specific Diagnoses in Sports Medicine

Sports Concussion

Concussions are mild traumatic brain injuries sustained as a result of a direct blow to the head or forces transmitted through the head and neck.¹⁸² Concussion occurs in all sports but is most common in contact and collision sports. Symptoms can present immediately or evolve over minutes to hours and often include headache, behavioral changes, cognitive impairment, and sleep disturbance (Box 40.8). These symptoms are likely the result of a

• BOX 40.8 Symptoms Reported After Concussion

- Headache
- Neck pain
- Nausea and/or vomiting
- Vision changes
- Balance problems
- Sensitivity to light
- Sensitivity to noise
- Feeling “slowed down”
- Feeling “in a fog”
- “Don’t feel right”
- Difficulty concentrating
- Difficulty remembering
- Fatigue
- Confusion
- Drowsiness
- Trouble falling asleep
- Changes in emotionality
- Irritability
- Sadness
- Nervousness or anxiety

functional disturbance of the brain rather than a macroscopic structural injury. However, possible axonal injury or blood flow changes may occur on a microscopic level not able to be seen with standard imaging protocols. It is vital to recognize that concussions often do not result in a loss of consciousness. Concussion symptoms are generally transient, although persisting symptoms can occur.

Pathophysiologically, the concussed brain exhibits increased metabolic needs combined with decreased cerebral blood flow.¹⁶ This mismatch of “supply and demand” creates a state of tissue vulnerability. Second-impact syndrome is a rare and catastrophic outcome related to concussion that might be caused by this mismatch. This phenomenon is seen when a youth athlete receives a second impact before the symptoms of the first concussion have subsided. These rare injuries have been reported only in youth athletes and result in severe brain injury and even death from malignant cerebral swelling.

The most vital component of concussion care is the prompt recognition of the injury itself. Athletes, coaches, trainers, and family members should have some degree of education regarding the most common signs and symptoms. Many states mandate annual coaching education on concussion as part of their concussion laws. An athlete with a suspected concussion should be removed from play immediately and evaluated on the sideline. The Sport Concussion Assessment Tool 6 is commonly used as a sideline tool and is available for free online.¹⁸² The current consensus is that all athletes, regardless of age or competition level, should not RTP on the same day as a concussion. Imaging is initially reserved for situations in which an intracranial abnormality is suspected, including prolonged loss of consciousness, focal neurological deficit, or progressive decline in neurological status. In athletes with prolonged or severe symptoms, magnetic resonance imaging (MRI) can be used to evaluate for other intracranial abnormalities. Newer imaging protocols, such as functional and diffusion-weighted MRI, positron emission tomography, and diffusion tensor imaging, are emerging but are not yet part of current clinical concussion management. Neuropsychological (NP) testing is a useful component in the management of concussions. Cognitive and somatic symptoms typically improve in parallel but often completely resolve at different times. As such, NP testing can give added information that cannot be gained in a typical clinical setting. Neuropsychologists are best suited to interpret these data, but in situations where no neuropsychologist is available, other medical professionals can be trained to perform the interpretation.⁶⁶ However, NP testing is only one component of concussion management, and the decision to RTP is ultimately a medical one.

In the first several days to weeks after an injury, athletes with concussions are managed with relative physical and mental rest. This includes limiting activities that require concentration and attention, such as scholastic activity, video games, computer use, and text messaging. These activities often exacerbate the athlete’s symptoms and can prolong the final resolution of the event. Youth athletes may require individualized schoolwork adaptations and the cooperation of school administration (e.g., extended test-taking time, tutors, limited class schedules). Section 504 of the Americans with Disabilities Act, part of a federal law that protects students with disabilities, is often used to create plans to help concussed athletes continue to participate in school. Data support that athletes with prolonged recoveries are safe to begin subthreshold exercise programs, which may speed recovery. The optimal timing and type of exercise are unclear, but subsymptom threshold aerobic exercise

TABLE 40.6 Return-to-Play Protocol

Rehabilitation Stage	Functional Exercise at Each Stage of Rehabilitation	Objective
1. Symptom-limited activity	Daily activity that does not aggravate symptoms	Gradual reintroduction of work and school
2. Light aerobic exercise	Walking, swimming, or stationary cycling; low intensity, no resistance training	Increase heart rate
3. Sport-specific exercise	Drills, no head impact activities	Add movement to test coordination and more complex motor skills
4. Noncontact training drills	Progression to more complex training drills, add resistance exercise	Exercise, coordination, and cognitive load
5. Full-contact practice	Participation in normal training	Restore confidence and assess functional skills
6. Return to play	Normal game play	

Modified from Patricios JS, Schneider KJ, Dvorak J, et al. Consensus statement on concussion in sport: the 6th International Conference on Concussion in Sport-Amsterdam, October 2022. *Br J Sports Med.* 2023;57(11):695-711.

can be initiated within 48 hours of injury. Vestibular rehabilitation has been shown to aid athletes suffering from dizziness, vertigo, or impaired balance after concussion. Optimal timing and duration of these programs are not known, but they should be utilized at the treating provider's discretion.²⁰⁵ Prognosis after sports-related concussion is typically excellent, with 80% to 90% of athletes free of symptoms within 7 to 10 days.¹⁵⁰ Children and adolescents are notable exceptions and may have a longer recovery time.¹⁵¹ RTP decisions can be challenging and must be made by practitioners familiar with concussion management. No athlete should RTP until all the symptoms of the concussion have resolved, and the athlete has returned to baseline. Once the athlete is asymptomatic at rest, a stepwise approach RTP protocol must be completed before competition is resumed (Table 40.6). The athlete will need a 24-hour symptom-free period before advancing to the next step in the protocol. No athlete should be returned to competition while still symptomatic.

Athletes with repeated concussions must be evaluated on an individual basis. These athletes may have a prolonged recovery and may have higher symptom severity.^{44,45,47,213} An increasing body of literature suggests concussions are an independent risk factor for long-term deficits in executive function and processing speed.^{45,92} The phenomenon of athletes suffering increasingly severe concussions with decreasing impact severity should raise significant concerns for the physician. These athletes may need to consider giving up contact sports and may not be cleared to participate.³¹ Chronic traumatic encephalopathy (CTE) is a disease thought to be caused by repeated concussive or subconcussive blows to the head. Pathologically, CTE involves the deposition of tau protein and amyloid plaques within the brain.¹⁵⁴ This pattern and the accompanying syndrome are distinct from other tauopathies and other neurodegenerative diseases. A formal consensus for neuropathological diagnosis has been developed for CTE, although diagnosis cannot be confirmed in living persons.¹⁵³ There have been cases reported in American football, martial arts, soccer, boxing, and wrestling, and it has mainly been confirmed in professional athletes, although confirmed cases have been seen in teenagers. CTE is not a clinical diagnosis.¹⁸² Traumatic encephalopathy syndrome (TES) has been proposed to describe such symptoms as mood changes, memory deficits, impulse control problems, confusion, and dementia in individuals who have

been exposed to repeated head trauma. CTE remains an area of debate and an opportunity for further research.⁷ CTE pathology has appeared most robustly in brain banks of former professional athletes with high exposure to repetitive head impacts and has been relatively uncommon in community samples. Despite the increasing numbers of confirmed cases of CTE, we cannot fully elucidate the threshold (i.e., number of head impacts or years of exposure) for development of TES or CTE or the potential intrinsic, individual vulnerability to develop the condition. There does seem to be a dose-response phenomenon involved in CTE, with more cases confirmed in former professional athletes and fewer complaints noted in former high school athletes as they age.^{56,159}

Concussion is increasingly recognized as a public health concern. Physiatrists, with a unique expertise in brain injury, sports medicine, and the complexities of managing the expectations of multiple involved parties, are poised to help shape public policy. The Lystedt law was the first concussion management law, passed by the State of Washington in 2009. It requires that all youth athletes with suspected concussion be removed from play, evaluated, and cleared by a medical professional before returning to play. All 50 states have since passed similar laws, and sports medicine physicians must practice in accordance with those laws.

Stingers

The stinger, sometimes termed a burner, is one of the most common but least understood peripheral nerve injuries in sports. Stingers are nerve injuries that occur within the peripheral neural axis at a specific but variable point from the nerve root to the brachial plexus. Because of the transient nature of symptoms and playing time considerations, stingers often go unreported to coaching and medical staff. Thus the true incidence of stingers is unknown; however, it is estimated that more than 50% of high school, collegiate football, and National Football League players sustain a stinger each year.²² Stingers can result from either a tensile (stretch) or compressive injury to the cervical nerve root or brachial plexus. Of the two, the cervical nerve root appears to be at greater risk for both tensile and compressive injury than the brachial plexus. However, the literature slightly favors brachial plexus tensile overload¹ over nerve root stretch,²¹⁵ nerve root compression,^{188,200,237} and direct brachial

plexus compression^{57,144} as the primary mechanism of injury in stingers. The pathomechanics of each injury vary by the age and experience of the athlete and by specific sport. Tensile injuries typically occur in younger athletes with less experience who have weaker neck and shoulder girdle musculature, leaving them vulnerable to forceful contralateral neck lateral bending with ipsilateral shoulder and arm depression. Cervical root compression is likely to occur in the older, stronger, and more experienced athlete during forceful cervical extension and rotation, narrowing the neuroforamen. This is likely related to early spondylitic changes in the cervical spine. This scenario is more commonly seen in professional football defensive backs and offensive linemen. Least commonly, a compressive blow to the brachial plexus, typically at the level of the supraclavicular fossa, can result from equipment issues such as padding.

The patient with a stinger classically presents with sudden onset of a lancinating, burning pain in one upper limb after a traumatic event. The symptoms typically follow a single dermatomal distribution, most commonly in a C5, C6, or C7 pattern. The pain usually lasts seconds to minutes, with the sensory disturbance usually resolving quickly; in contrast, weakness can be more persistent. Simultaneous, bilateral stingers are very uncommon; therefore any athlete with bilateral upper limb paresthesias or dysesthesias should be evaluated for a SCI injury and/or transient quadriplegia. With the first occurrence of a stinger, symptoms generally resolve within minutes to hours, and no medical attention is sought. With each subsequent recurrence, more distinct neurological sequelae might result, including persistent motor weakness. Motor impairment particularly in the deltoid and biceps (i.e., C5 myotome) is the more common residual neurological symptom of an athlete with a stinger.⁹⁶

The initial sideline evaluation includes identification of the mechanism of injury and the precise distribution and duration of symptoms. The sideline physical examination includes cervical active ROM to assess pain provocation and rigidity, palpation for tenderness, and a detailed neurological examination with close attention to the C5-C7 myotomes. The duration of any given episode can vary from seconds to hours, much less commonly lasting for days or longer. Serial assessments are critical for RTP and management decisions. If symptoms progressively worsen over the first few days or if weakness persists more than 10 to 14 days, additional ancillary testing and specialty consultation would be indicated to precisely determine the site of injury and the degree of axonal damage. If the athlete's symptoms resolve very quickly (within seconds to minutes) and physical examination demonstrates no abnormality (including full pain-free cervical ROM), then no additional testing may be necessary. Persistent symptoms or signs and recurrent stingers are indications for further workup.

Although plain radiographs of the cervical spine provide limited information in this setting, they can reveal clues to pathoanatomy contributing to the symptoms. A higher incidence of degenerative changes (creating central canal and neuroforaminal stenosis concomitantly) is noted in more experienced athletes who sustain stingers.¹³³ Levitz et al.¹³³ studied a cohort of football players with chronic stingers and demonstrated that 93% had significant cervical disc disease or neural foraminal narrowing. Advanced imaging with MRI or computed tomography is indicated if the athlete experiences severe unrelenting pain, persistent symptoms, or abnormal neurological findings on examination. Electrodiagnostic (EDX) testing is complementary to imaging studies. EDX can provide quantification of neural dysfunction and can help to determine the site of nerve injury. EDX is most

useful in the athlete with persistent or progressive weakness beyond 2 weeks postinjury or in the setting of recurrent stingers. As with other nerve injuries, 7 to 10 days may be required for the axonal damage to manifest on needle electromyography (EMG), so this is not a test used immediately postinjury.

Treatment of stingers includes pain control, strengthening, and rehabilitation of postural faults and muscle imbalances that may have contributed to the athlete's risk of sustaining the stinger. Persistent symptoms can be treated with fluoroscopically guided epidural steroid injections, surgical decompression of neuroforaminal stenosis, or even anterior cervical discectomy and fusion. Equipment modifications should be considered for prevention of recurrent injury, although research is lacking to support their effectiveness.⁹⁶

RTP decisions for athletes with stingers are made on a case-by-case basis. The following is an approach based on a combination of clinical experience, neurophysiological principles, and extrapolation of information from other peripheral nerve injuries. After an initial stinger, if full recovery is demonstrated within 5 minutes, return to same-game competition is allowed.²¹⁹ If full recovery occurs within 1 week after the initial stinger, then return to competition that next week may be considered. A limited rehabilitation program to address postural dysfunction and relative weaknesses should be prescribed during the in-season strength and conditioning program, followed by a more comprehensive program in the off-season. If the athlete has sustained recurrent stingers, further evaluation with cervical MRI should be considered to assess for structural abnormality before considering RTP. If more than three stingers occur in a season, ending the season should be considered, especially if there is significant weakness, axonopathy on EMG, or focal disc herniation or foraminal stenosis noted on MRI.⁹⁷

Exercise-Induced Bronchospasm

EIB describes airway narrowing that occurs in association with exercise. EIB can be present with chronic asthma but generally is a separate entity. The prevalence of EIB is significantly higher in athletes than the general population. A diagnosis based solely on clinical symptoms is often inaccurate, and pulmonary function testing and consultation with an asthma specialist are important to definitively diagnose and optimally treat EIB.²⁰¹ Symptoms can include breathlessness with or without coughing and wheezing or even solely the vaguer symptom of decreased performance during vigorous endurance training or competition. Symptoms usually peak 5 to 10 minutes after exercise ceases. Cross-country skiers and other winter sport athletes most often present with EIB, although runners are not far behind. EIB is much less common in swimmers. Cold, dry air is much more likely than warm, humid air to precipitate symptoms.

The mechanism of EIB is not fully understood but is certainly different from that of chronic asthma, which is caused by inflammation resulting from overly reactive airways to inhaled stimulants. EIB is thought to result from water loss and cooling in the airway that occurs with hyperventilation, which subsequently trigger bronchoconstriction.³³ Several diagnostic testing options are available when EIB is suspected by history, but the physical examination is normal (Box 40.9).³³ Once a diagnosis of EIB is made, treatment is multifaceted. An adequate warmup has been demonstrated to reduce the severity of EIB. Some recommend inducing a refractory period by briefly precipitating symptoms with short, vigorous bursts of exercise lasting 15 to 20 minutes

• BOX 40.9 Diagnostic Testing for Exercise-Induced Bronchospasm

- Spirometry
- Formal pulmonary function tests
- Challenge testing
- Eucapnic voluntary hyperventilation challenge
 - Breathe in 4.5% CO₂ for eucapnia, then measure FEV₁
 - Any decrease in FEV₁ is diagnostic for EIB
- Exercise challenge in the field or laboratory
 - Useful because it simulates the sport-specific temperature/environment
- Methacholine challenge
 - Good to classify the severity of asthma
- Not specific for EIB

EIB, Exercise-induced bronchospasm; FEV₁, forced expiratory volume in 1 second.

before the competitive endurance event. The mainstay of pharmacological therapy is a SABA 15 minutes before exercise, although daily use will lead to tolerance and decreased efficacy.⁴⁰ If SABAs are ineffective, are being used daily, or the athlete becomes tolerant, a controller agent such as a leukotriene antagonist (LTRA) or ICS is indicated.^{8,34} LTRAs do not lead to tolerance, but efficacy is limited to approximately 50% of users. ICSs are first line for athletes with concomitant chronic asthma. Long-acting beta agonists should be used less than three times per week and never used as monotherapy because of increased mortality and morbidity.^{179,239}

Anemia

Common causes of anemia in the athlete include iron deficiency anemia (IDA), physiological anemia (pseudanemia), and foot-strike hemolysis.

IDA is most common in menstruating females, and female athletes can be more prone to developing it. The etiology is either blood loss or poor iron intake. Many athletes consume restrictive diets that can have too little iron to meet daily needs. However, a complete history and physical examination are still important to evaluate for nonathletic causes, such as significant gastrointestinal or genitourinary blood losses. Usually, a serological workup is diagnostic and includes complete blood count (CBC), serum ferritin, and total iron binding capacity (TIBC). The CBC will show a microcytic anemia.¹²⁵ Serum ferritin levels that are less than 30 ng/mL in athletes are considered suggestive of IDA.⁷³ The TIBC will be elevated. If IDA is diagnosed, a trial of oral iron supplementation (typically ferrous sulfate or ferrous gluconate, 325 mg three times daily) is undertaken.⁴³ Iron is best absorbed in an acidic environment, so it is concomitantly administered with vitamin C usually for a 2- to 3-month course.

Physiological anemia is considered a pseudanemia seen commonly in endurance athletes. Endurance athletes tend to have a lower hemoglobin concentration than the general population because of plasma volume expansion with a dilutional effect. It is an adaptation to exercise and is not considered to inhibit athletic performance. It generally normalizes with training cessation of 3 to 5 days. If IDA is ruled out, there is no necessary treatment.

Foot-strike hemolysis refers to red blood cell destruction in the feet from running impact. However, intravascular hemolysis is seen in swimmers, cyclists, and runners, but whether actual mechanical red blood cell trauma is the source is questionable. Possible reasons are intramuscular destruction, osmotic stress, and membrane lipid

peroxidation caused by free radicals released by activated leukocytes. Intravascular hemolysis can even be regarded as a physiological means to provide heme and proteins for muscle growth in athletes.¹⁹⁹ In general, the hemolysis is mild, and treatment is rarely required.

Specific Populations

Females in Sports

Until puberty, males and females are similar in body size, shape, and physiology.²⁰ On average, females have a larger surface area to mass ratio, lower bone mass, and a wider, shallower pelvis than males. Because of these differences, females may tolerate heat better than males, may be more prone to osteoporosis, and may develop knee problems more frequently.²⁰ The following discussion highlights one issue common to the female athlete, previously widely regarded as the female athlete triad, though now more often referred to as relative energy deficiency in sport (RED-S).

The female athlete triad was originally described as a constellation of conditions including low energy availability (LEA) (with or without disordered eating), menstrual dysfunction, and low bone mineral density (BMD).^{52,175} The definition of the triad does not require the simultaneous occurrence of all three components. Each facet, in isolation or combination, needs to be considered and treated.¹⁷¹ Historically, laboratory studies have standardized energy availability (EA) to kilogram of fat-free mass (FFM), with EA values of 45 and 30 kcal/kg FFM/day as the physiological optimum and the LEA threshold, respectively.¹⁴⁰ As LEA occurs, diminished athletic performance often results, and a vicious cycle develops in which psychological distress and depression can develop.¹⁷⁰ LEA can occur in those who eat insufficient calories for the amount of calories burned, as well as in people with disordered eating, such as anorexia or bulimia. The incidence rates of disordered eating are higher in female athletes than in other populations.^{20,90} An age-matched control study of elite Norwegian athletes found incidence rates of 20% in female athletes, compared with 9% in female nonathletes, 8% in male athletes, and 0.5% in male nonathletes.²¹⁸ Rates are even higher among athletes in sports emphasizing leanness, such as gymnastics, figure skating, dancing, distance running, diving, and swimming.^{13,20,170,171}

Female athletes are three times as likely to develop primary or secondary amenorrhea as their nonathletic counterparts.¹⁹¹ Menstrual irregularities range from reduction in luteal phase length and suppression of luteal function to amenorrhea.²⁰ Females with athletic amenorrhea present with a reduction in luteinizing hormone pulse frequencies, which depend on EA.¹⁷⁰ Dietary supplementation (i.e., providing adequate energy sources) can prevent or eliminate menstrual disturbances.^{20,90} Although an LEA threshold of 30 kcal/kg FFM/day has been reported, prospective studies manipulating energy intake and energy expenditure in asymptomatic subjects found that menstrual disturbance was more affected by the magnitude of energy deficit from baseline rather than at a specific threshold.^{137,247}

Skeletal demineralization is the third component of the classic triad and can lead to premature osteoporosis. The prevalence of osteopenia in female athletes has been reported to be 22% to 50%, whereas the prevalence of osteoporosis remains relatively low.¹¹⁸ Bone undergoes continual resorption and formation, a process called remodeling. Hormonal status, weight-bearing physical activity, and dietary intake (particularly calcium) influence remodeling. Stress fractures pose a health risk to amenorrheic athletes as a result of reduced bone mass. The relative risk for stress fractures is two to

four times greater for amenorrheic athletes compared with those with regular menses.¹⁷¹ Peak bone mass is achieved during the first 3 decades of life, with Tanner stages II and III (early to middle puberty) as the maturational stage in which physical activity has the strongest impact on bone.¹⁷¹ Young athletes with primary amenorrhea might be unable to build bone during formative years, with the end result being that peak bone mass might never be achieved.²⁰ Dual-energy x-ray absorptiometry is the most widely used and validated technique to assess BMD.^{115,171} The World Health Organization defines osteoporosis as a BMD greater than 2.5 standard deviations (SD) below the youthful mean (T-score ≤ -2.5 SD).¹¹⁵ Osteopenia is defined as a BMD between 1 and 2.5 SD below the youthful mean (T-score 1.02.5) (Table 40.7).¹¹⁵ Notably, normative data for adolescent females are not available to definitively define pathological bone loss in this group; hence Z-scores are best used in the skeletally immature athlete because the Z-score is age matched, whereas the T-score is compared with a 35-year-old person.^{13,20} RCTs have shown that oral contraceptive pills might increase lumbar and total body BMD in patients with hypothalamic amenorrhea.^{100,235} In addition, 1200 to 1500 mg of calcium and 400 to 800 IUs of vitamin D/day are recommended.⁹⁰

Female athletes presenting with features of the female athlete triad and/or relative energy deficiency require appropriate workup and referrals (Fig. 40.8). Although referrals to a nutritionist, psychologist/therapist, and other medical specialists, such as endocrinologists, cardiologists, and gastroenterologists, may be necessary, a primary treatment focus should be to restore energy balance by improving caloric intake and potentially decreasing energy expenditure (limiting aerobic exercise). Finally, pregnancy should be excluded in any amenorrheic female of child-bearing age. The IOC published a consensus statement in 2014 titled, “Beyond the Female Athlete Triad: Relative Energy Deficiency in Sport (RED-S),” and an update in 2018, proposing that LEA leads to impairments beyond bone health and menstrual function, including in metabolic rate, endocrine function, immunity, protein synthesis, and cardiovascular health (Table 40.8).^{163,164} In addition, RED-S expands the consequences of LEA to include male athletes—identifying at-risk populations, including cyclists, rowers, runners, jockeys, and weight-class athletes—and para-athletes, who may be at risk of nutrient deficiency and disuse osteopenia/osteoporosis and may have aberrant movements or asymmetrical gait patterns resulting

in increased energy expenditure. In place of menstrual irregularities, males are at risk for hypogonadotropic hypogonadism.²²²

Pediatric and Adolescent Athletes

It has been estimated that over 60 million youth athletes participate in organized sports in the United States.¹³⁸ As athletic involvement increases, the incidence of sports-related injuries also increases.^{128,214} Physically active adults had better standardized fitness scores as children, suggesting that an active lifestyle as a child promotes continued lifelong physical fitness.⁵⁴ Physical activity in children and adolescents prevents osteoporosis, improves self-esteem, and reduces anxiety and depression.^{17,37,138} Furthermore, there is a positive relationship between physical fitness and academic achievement in English and mathematics.⁴¹ High school athletes are more likely to engage in positive health behaviors and less likely to engage in negative health behaviors compared with nonathletes.¹⁸¹ Similar findings in regard to social connectedness, self-esteem, depression, and suicide have held true in collegiate athletes as well.^{6,195}

When advising pediatric athletes about their training program, the physiological and anatomical differences between children and adults must be considered. Strength training has been historically discouraged in the pediatric population because of concerns of growth disturbance and potential for other injuries.¹²⁸ However, growing evidence suggests this may be inaccurate; in fact, well-designed strength programs likely provide performance and health benefits in the youth athlete.⁷² Resistance training programs should be individualized to the athlete with consideration of the athlete's technical competency and health history. Children may safely benefit from resistance training programs as young as 5 to 7 years old. Most youth athletes can perform resistance training two to three times weekly with regular participation leading to benefits in power, fitness, and performance.⁷²

The pediatric skeletal system is distinguished from that of adults by the presence of the active growth plate, or physis. The physis is situated between the epiphysis and metaphysis. Two types of epiphyses exist: traction and pressure.²⁹ The traction epiphysis, also known as the apophysis, is the point of attachment of a tendon to bone.¹ Because apophyses contribute to bone shape but not bone length, acute or chronic injuries involving apophyses are not usually associated with growth disturbance. Osgood-Schlatter disease, Sever

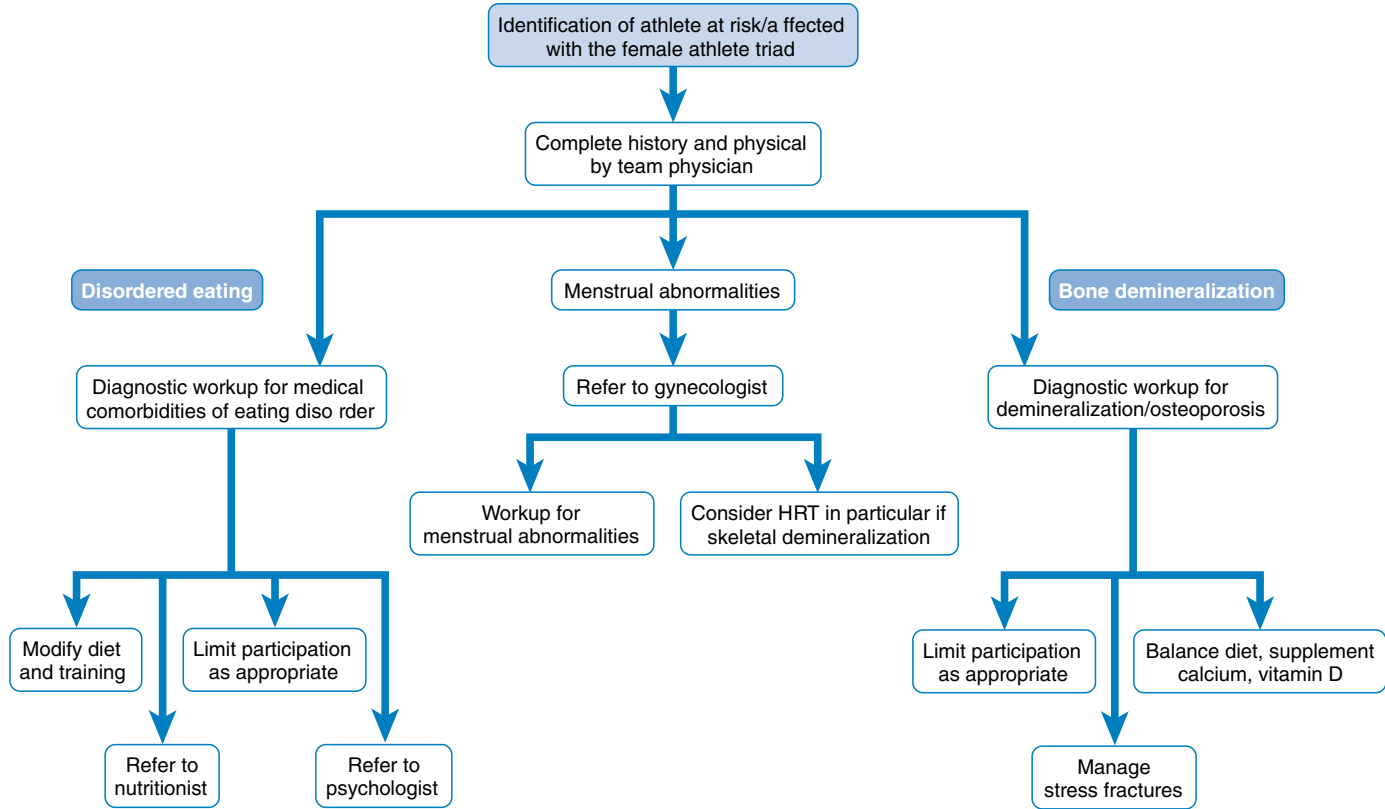
TABLE 40.7 Current Recommendations for the Diagnosis of Low Bone Mineral Density and Osteoporosis in Premenopausal and Postmenopausal Females and Young Athletes

	World Health Organization	International Society of Clinical Densitometry	International Olympic Committee
Targeted population	Postmenopausal females	Premenopausal females	Young, premenopausal athletes
Terminology	Osteopenia	BMD below expected range for age	BMD below expected range for age
Proposed cutoff score	T score: -1 to -2.5	>20 years of age: Z-score ^a ≤ -2	>20 years of age: Z- or T-score ≤ -1 should be of concern
Terminology	Osteoporosis	Low BMD for chronologic age or below expected range for age	Osteoporosis in athlete with amenorrhea
Proposed cutoff score	T-score: ≤ -2.5	<20 years of age: Z-score ^a ≤ -2	>20 years of age: Z-score ^a -2.5

BMD, Bone mineral density.

^aZ-score and T-score may be similar in females older than 20 years.

Modified from Beals KA, Meyer NL. Female athlete triad update. *Clin Sports Med*. 2007;26:69-89.



• **Fig. 40.8** Overview of the management of the female athlete triad. HRT, Hormone-replacement therapy. (Modified from Borg-Stein J, Dugan SA, Solomon JL. Special considerations in the female athlete. In: Frontera WR, Herring SA, Micheli LJ, et al., eds. *Clinical Sports Medicine: Medical Management and Rehabilitation*. Saunders; 2007.)

TABLE 40.8 Health and Performance Consequences of RED-S

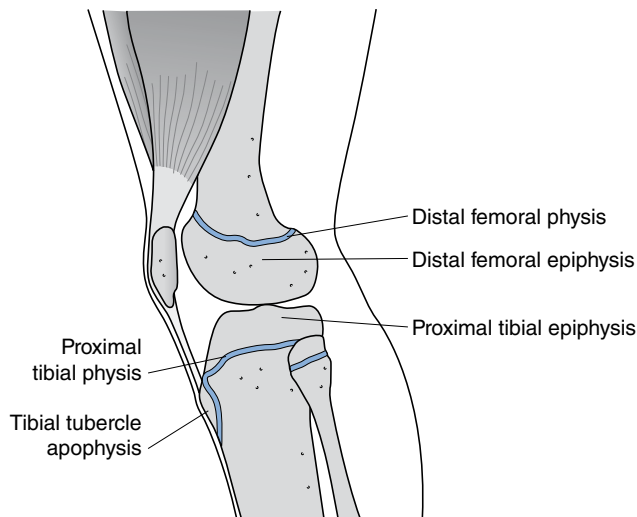
Health Consequences of RED-S	Performance Effects of RED-S
Menstrual function	Decreased endurance
Bone health	Increased injury risk
Endocrine	Decreased training response
Metabolic	Impaired judgment
Hematologic	Decreased coordination
Growth/development	Decreased concentration
Psychological	Irritability
Cardiovascular	Depression
Gastrointestinal	Decreased glycogen stores
Immunologic	Decreased muscle strength

RED-S, Relative energy deficiency in sport. Modified from Mountjoy M, Sundgot-Borgen J, Burke L, et al, International Olympic Committee (IOC) consensus statement on relative energy deficiency in sport (RED-S): 2018 update. *Int J Sport Nutr Exerc Metab*. 2018;28:316-331.

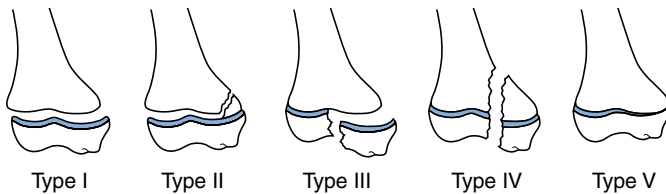
disease, and medial epicondylopathy are examples.^{1,29} Pressure epiphyses are found at the end of long bones such as the distal femur and proximal tibia (Fig. 40.9). In contrast to apophyseal injuries, injuries to active pressure epiphysis can result in limb-length discrepancies or angular deformities.^{29,212} The most common cause of acute injury to the pressure epiphysis is a fall. Acute physal injuries are widely described using the system by Salter and Harris (Fig. 40.10). Chronic injuries result from ongoing stress to the physis, as is seen in distal radial injuries in young gymnasts (Fig. 40.11).²⁹

Older Athletes

Many older adults remain physically active. Popular activities among seniors include gardening, dancing, golf, hunting, fishing, woodworking, tennis, bowling, biking, and swimming. Masters athletes have achieved impressive improvements in peak exercise performance. In fact, athletes older than 70 years of age have surpassed the winning time at the first modern Olympic Games held in Athens in 1896 in running events from 100 m to the marathon.²²¹ Over the past 2 decades, the benefits of exercise and the costs of a sedentary lifestyle have become well known, with the American College of Sports Medicine trademarking “Exercise is Medicine” for their global health initiative promoting physician-prescribed exercise.⁵⁹ This is true for older adults as well, for whom the US Department of Health and Human Services recommends 150 minutes or more of aerobic exercise per week, strength training of all major muscle groups twice per week, and balance training (Table 40.9).²²⁸ There is strong evidence that



• **Fig. 40.9** Physes of the distal femur and proximal tibia. (Modified from Caine D, DiFiori J, Maffulli N. Physeal injuries in children' and youth sports: reasons for concern? *Br J Sports Med.* 2006;40:749-760.)



• **Fig. 40.10** Growth plate injuries as described by Salter and Harris. (Modified from Caine D, DiFiori J, Maffulli N. Physeal injuries in children's and youth sports: reasons for concern? *Br J Sports Med.* 2006;40:749-760.)



• **Fig. 40.11** Radiograph of a chronic distal radius injury resulting from ongoing stress to the physis.

TABLE 40.9

US Department of Health and Human Services 2018 Physical Activity Guidelines

	Aerobic Exercise	Resistance
Children (6–17 yr)	60+ min total of aerobic and strength training daily Include vigorous aerobic activity at least 3 days/week <i>Options: running, hopping skipping, jumping rope, swimming, dancing, cycling</i>	60+ min total of aerobic and strength training daily Include muscle strengthening at least 3 days/week <i>Options: playground equipment, climbing, tug of war, free weights, resistance bands, gymnastics</i> Include bone strengthening at least 3 days/week <i>Options: running, jumping, basketball, tennis, hopscotch, sports involving rapid change in direction</i>
Adults	150–300+ min/week of moderate intensity <i>Options: walking > 2.5 mph, recreational swimming, cycling < 10 mph, doubles tennis, yoga, ballroom or line dancing, yard work, water aerobics</i> 75–150 min/week of vigorous intensity <i>Options: running, swimming laps, singles tennis, dancing, cycling > 10 mph, jumping rope, hiking uphill or with a pack, high intensity interval training, exercise classes</i>	Strength training that involves all major muscle groups ≥2 days/week
Older Adults (65+ yr)	Same as adults <i>Options: walking, dancing, swimming, water aerobics, jogging, running, yoga, cycling, yard work, tennis, basketball, golf</i> + Incorporation of balance training + Effort should be relative to fitness level and chronic conditions	Same as adults <i>Options: exercise bands, weight machines, free weights, body-weight exercises, gardening, tai chi, yoga, carrying groceries</i> + Incorporation of balance training + Effort should be relative to fitness level and chronic conditions

Modified from US Department of Health and Human Services. *Physical Activity Guidelines for Americans*. 2nd ed. HHS; 2018.

physical activity is associated with lower incidence rates and decreased symptoms in many afflictions associated with aging, including psychiatric illness, neurological disease, metabolic disease, cardiovascular disease, diabetes and obesity, pulmonary disease, musculoskeletal disorders, and cancer, and confers an overall reduction in mortality.^{109,184} A 2018 study of muscle biopsies in lifelong exercisers older than 70 years concluded that exercise can maintain the muscle tissue phenotype (capillarization and aerobic enzymes) of adults 50 years younger.⁸⁹ Even short bouts of exercise may boost immune function, decreasing the likelihood of acute infection.¹³⁰ One theory for the pervasive benefit of exercise is an associated reduction in oxidative stress and inflammation, both of which are known to increase with aging.⁴² A 2015 review provided a summary of evidence-based exercise recommendations for 26 chronic diseases and included proposed mechanisms of effect.¹⁸⁴

Multiple studies have confirmed the cognitive benefits of exercise in various older populations, including but not limited to groups with Alzheimer disease, Parkinson disease, and sedentary lifestyles.^{11,108,176} In one survey, 68% of athletic seniors regarded their quality of life as much better than that of their sedentary friends.²⁰⁹ A systematic review of RCTs of exercise training in elderly, institutionalized adults found strong evidence that exercise increases strength and mobility in this population.²⁰² Moderate evidence was found for an effect on ROM, and contradictory evidence was seen regarding gait, activities of daily living, balance, and endurance. The authors concluded that more research is necessary to know the extent of possible advantages of exercise in this population. Notably, there were no ill effects reported as a result of exercise.²⁰² Exercise recommendations should be individualized to the athlete and should be prescriptive, similar to medication, detailing dose, intensity, type of exercise, and frequency. Recommended exercises may be adapted based on the goal outcome (e.g., aerobic exercise in the treatment of cardiovascular disease or a focus on weight-bearing resistance exercise in the treatment of osteoporosis).¹⁰⁹

Transgender Athlete

Elite-level sport participation by transgender athletes has received international attention in recent years. The IOC and NCAA have adopted new guidelines around decisions of inclusion and fairness in this population.^{107,168} As stated by the IOC, “Every person has the right to practice sport without discrimination and in a way that respects their health, safety, and dignity. At the same time, the credibility of competitive sport, and particularly high-level organized sporting competitions, relies on a level playing field, where no athlete has an unfair and disproportionate advantage over the rest.” It has been recommended that sport-specific guidelines for transgender athlete participation be developed by national and international sport federations with this framework in mind.

It should not be assumed that any one person has an unfair competitive advantage based upon sex alone. However, it is known that at the elite level, in sports involving strength, speed, power, or endurance, elite males generally outperform elite females. This is largely because of the effects of testosterone.¹⁷² Before puberty, athletic performance is similar between the sexes. Gender-affirming care includes medical and surgical treatments that allow an individual to align one's outward self with an inward gender identity. Gonadotropin hormone-releasing hormone (GnRH) agonists may be used to delay gender-incongruent puberty in adolescents. Lower BMD is a potential consequence of GnRH agonist treatment, though BMD

seems to rebound once hormonal treatment is initiated. In adolescents receiving GnRH agonists who have not yet started hormonal treatment, there is a theoretic but not definitive increased risk of Salter-Harris type fractures.¹⁵² Testosterone may be used as a masculinization treatment in transgender men and has been associated with increased hemoglobin concentration, lean body mass, and muscle cross-sectional area, along with reduced body fat.¹⁷² In terms of athletic performance, after at least 3 years of testosterone treatment, transgender men may be able to reach competitive levels within the range of cisgender men, but there is no evidence to suggest transgender men outperform cisgender men at the elite level. Feminization can be achieved with 17-beta-estradiol treatment, with peak effects occurring after 2 to 3 years of treatment initiation. GnRH agonists such as spironolactone or cyproterone acetate are often used as well to suppress testosterone levels. This treatment is associated with decreased hemoglobin concentration, reduced muscle mass, and increased body fat.¹⁷² In regard to athletic performance, data are unfortunately lacking. However, studies have suggested transgender women, compared with cisgender counterparts, maintain a performance advantage in strength and endurance at 1 year posttreatment.¹⁷² Less data are available at 2 to 4 years, but after this duration of treatment athletic performance in transgender women more closely approximates that of cisgender women.⁴⁰ Of note, there are some attributes that remain unaffected by gender-affirming hormone therapy, including height and limb length.

Transgender individuals face several barriers to athletic participation and overall are less physically active than cisgender counterparts. Lack of inclusive and comfortable environments likely plays a large role. The locker room setting has been identified as an environment that can lead to psychosocial distress for the transgender athlete.^{152,172} Comorbidities, including high rates of suicidality, depression, anxiety, and eating disorders, may also play a role in one's ability to engage in sports. Furthermore, legislation in some states restricts access to organized sports to sex assigned at birth. Care of the transgender athlete requires a thoughtful and evidenced-based approach. The sports medicine physician should be familiar with the physiological effects of gender-affirming treatments and stay up to date on the ever-evolving, sport-specific guidelines surrounding transgender participation.

Adaptive Sports Medicine

Over the past few decades, athletes with impairments have been one of the groups with the most rapidly increasing levels of sports participation. The combination of an emphasis on the population to maintain physical fitness along with medical advances has helped grow the population of persons with a disability to participate in competitive and recreational sporting activities. According to the US Census Bureau, approximately 45 million people living in the United States had a disability in 2021.²²⁷ Although the number of people with a disability who participate in sporting activities is increasing, there are still 56% of individuals with a disability who do not participate in sports in the United States.²²⁹ Despite the continued increase in the development of a number of recreational and competitive sports programs, approximately 60% of people with a disability do not participate in any regular physical activity or sports.²⁵⁰ Awareness and access remain the two most limiting factors for persons with disabilities to participate in athletics despite growing opportunities.²⁵⁰ Health care practitioners should make every effort to inform individuals of the multiple opportunities and encourage their safe participation

in recreational and competitive sports and exercise. There has been an increase in the numbers of athletes and sporting options in this population partly thanks to international organizations such as the IPC and affiliated national governing bodies (NGBs). Access to sports participation has been greatly improved by legislation that has advanced the rights of those with impairments. Growth is best exemplified through the phenomenal rise of the Paralympic Games as evidence by the increase in the number of participants from the 1960 Paralympics in Rome (400 athletes from 23 countries) to the 2012 London Games (4200 athletes from 164 countries).¹³⁹ Additionally, there were more countries that competed at the Beijing 2008 Paralympics (3951 athletes, 146 countries) than in the Munich 1972 Olympic Games. The degree of medical coverage has vastly improved, starting with the 2014 Sochi Winter Games, and is expected to reach an unprecedented level with the 2024 Paris Paralympic Games. There is a growing interest in and acceptance of sport for persons with a disability, and expansion of the Paralympics is most likely to continue.

This growth in participation has been paralleled with an increase in awareness of opportunities supported by the medical community. Sports participation among persons with impairments can have significant benefits in terms of general state of health, functionality, life skills, self-esteem, and overall quality of life.^{23,67,208,217,230} It is imperative that medical providers promote activity among patients with disabilities. However, the clinician should also be cognizant of the various complications associated with different disabilities and sports-specific activities to maximize patient safety and health. There are multiple benefits to encouraging persons with impairments to participate in recreational and competitive sports and exercise. Athletes with impairments demonstrate increased exercise endurance, muscle strength, cardiovascular efficiency, and flexibility. They also demonstrate improved balance and better motor skills compared with individuals with disabilities who do not participate in athletics. In addition to physical benefits, the psychological benefits of exercise include enhanced self-image, body awareness, motor development, mood, decrease in self-reported stress, pain, and depression, as well as a general increase in quality of life.¹⁰⁴ Athletes with disabilities have fewer cardiac risk factors and higher high-density lipoprotein cholesterol levels and are less likely to smoke cigarettes than those who are disabled and nonactive.^{53,78} Individuals with limb deficiencies who participate in athletics have improved proprioception and increased proficiency in the use of prosthetic devices.²⁴¹ In addition, wheelchair athletes with paraplegia are less likely to be hospitalized, have fewer pressure ulcers, and are less susceptible to infections than nonactive individuals with paraplegia.²¹⁷

Classifications

The IPC is the global governing body of the Paralympic movement and organizes the Summer and Winter Paralympic Games. It was founded on September 22, 1989, in Dusseldorf, Germany, and was preceded by the International Coordination Committee of World Sports Organizations for the Disabled. The IPC serves as the international federation for nine sports, including track and field, skiing (Alpine, Nordic, biathlon), and sled hockey.

Athletes with impairments are an inherently heterogeneous group with highly variable profiles of sports capacity, depending on type, location, and severity of the disability. Similar to the weight classes used in boxing and wrestling, classification systems have

been developed to maintain a measure of fairness in competition between those with both similar and different impairments. One of the first classification systems was used at the Stoke-Mandeville Games to avoid competition between athletes with dissimilar disadvantages resulting from their type of disability (e.g., a paraplegic individual competing against a tetraplegic individual). These classification systems have been formalized over the years by the IPC and its member federations and NGBs and are continually revised to reflect advancements in sports and medicine.

Although many systems exist and are used depending on the NGB, number of participants, and available resources, the common purpose is to determine eligibility for participation and ensure that athletes are not precluded from success in competition on the basis of their disability. To be eligible for Paralympic sports, an athlete must have 1 of 10 types of permanent impairment: ataxia, athetosis, hypertonia, leg length difference, limb deficiency, loss of muscle strength, loss of range of movement, short stature, low vision, or intellectual impairment. Athletes are placed more generally into six main disability categories: amputee, wheelchair, cerebral palsy, vision impaired, intellectual disability, or *les autres* (a French term meaning “the others,” referring to those who do not fit well within any other category).⁵⁰

Athletes are given a specific designation based on their disability category and functional ability. For example, athletes are classified based on different combinations of functional strength, truncal stability, ROM of arms and legs, use of a wheelchair, quality of ambulation and balance, and distribution and severity of muscle spasticity. Direct observation of the athlete during exercise and while participating in the activity is also often involved with some categories able to use more objective measures. This is of particular importance for evaluation of visual perception and field of view in the visually impaired category.

Because there are significant variances among the different sports, the sporting event also greatly influences the classification process. Within each sport, there are different types and severities of disability, which affect performance to varying levels. Depending on the sport, designations may be condensed into fewer classes based on a wider range of disability. In addition, some sports may have competition between athletes from different general disability categories who maintain similar functional levels. Team sports such as wheelchair basketball often incorporate a wide range of disabilities in direct competition via the use of point systems in which athletes with less disadvantageous levels of disability are assigned higher point values, and the team is not permitted to exceed a given total point value for its players in the game.

An athlete's classification is not permanently assigned and is always in evolution because there may be progressive worsening or improvement of the disease process, as well as the ability to adapt and overcome impairments. Athletes may change classifications throughout their life, depending on the system and changes in functionality experienced over time. However, this also means that athletes with impairments that are responsive to training may be classified in one category in one training state only to compete later in a different state of training, thus gaining a competitive advantage. Therefore it is imperative for the classifiers to be cognizant of this dynamic and evaluate for any intentional minimization of function during the evaluation process. These classification systems have also been criticized for a lack of peer-reviewed research supporting the limits and distinction of one designation from another and the use of objective criteria. In addition, the rapid advancement of technology and materials, particularly within the amputee category, has also given rise to significant concerns regarding the level

of competitive fairness given often-asymmetrical levels of resources among competitors. In light of these issues, the IPC has committed to a vision of classification that is evidence based and equitable to promote participation and competitive fairness. As such, continued assessment and revisions of these systems can be anticipated in years to come.

Adaptive Sports Equipment

There are several aspects to consider when the physician is evaluating the athlete with limb deficiency for adaptive equipment. Compared with the prosthetic device for everyday use, there are various modifications that should be considered for specialized use. Of particular importance is the weight of the prosthesis, particularly in sports where increased weight may affect speed. In addition, when prescribing an adaptive prosthesis, the clinician should consider alignment, prosthetic foot dynamics, shock absorption, and possible need for transverse rotation. Medical advances in rehabilitation and development of prosthetic devices have led to some controversy in competition against other adaptive athletes. Access to different types of technology and materials can differ greatly depending on the athlete's socioeconomic status and country of origin, creating the potential for significant competitive imbalance between similarly disabled and conditioned athletes. These advances have also been causes for controversy in competition against able-bodied athletes, with continued improvements in prosthesis weight and energy-storage capabilities being a potential source of a competitive imbalance. However, many of these comparisons do not take into account the increased energy cost of ambulation for the limb-deficient athlete. Therefore to fully assess the energy-storing advantages in the prosthetic limb, the additional energy needs for the limb-deficient athlete to ambulate with the prosthetic device must be considered. Determining the most appropriate methods for assessing and rectifying any potential competitive imbalance will likely be an area of considerable evolution in the near future.

There are also numerous adaptations that can be made to traditional wheelchairs, which can aid participation in various sports. Of note, adjustments can be made in seating systems, leg rests, wheel type, wheel position, wheel angulation, and length to enhance athletic performance in any specific sport. Many advances in manual wheelchairs trace their origins to wheelchair sports. Sporting wheelchair adjustments that were used for racing and basketball approximately 25 years ago have become integral to the manual wheelchairs that people currently use every day. Frame materials such as aluminum, titanium, and composite materials were all first introduced in sports wheelchairs. However, instead of solely improving sports performance, they help to increase the ability to independently perform activities and reduce repetitive strain injuries. In addition, the principles used in frame design of sports wheelchairs make them lighter and stiffer but ultimately faster and more responsive. In general, they also make the chairs easier to propel, allow for easier transport of the wheelchair in a vehicle, and reduce upper limb strain in athletes and nonathletes alike. These materials and principles are currently used widely in the design of ultralight manual wheelchairs. Given the vast modifications that can be made to a wheelchair to facilitate participating in sports, the full extent of discussion is beyond the scope of this chapter.

Injury Patterns in Adaptive Sports Medicine

Injury rates for athletes with disabilities are similar to those for athletes without disability; however, location of injuries appears

to be impairment and sport dependent. Lower limb injuries are more common in ambulatory athletes (visually impaired, amputee, cerebral palsy), whereas upper limb injuries are more frequent in athletes who use a wheelchair.⁷⁴ Athletes with lower limb deficiency are at risk for injuries in both the intact and residual limbs. The distal residual limb is often the site of skin trauma caused by prosthesis. Asymmetry is also commonly noted at the ipsilateral hip, as well as the pelvis and lumbar spine, as a result of increased hip power, pelvic rotation and obliquity, and spine lateral flexion and extension to improve clearance of the prosthesis during swing phase and to increase excursion and propulsion.¹⁹³ Such asymmetries may predispose athletes with these limb deficiencies to hip, sacroiliac, and lumbar spine pain.^{79,162} Alternatively, the intact limb may experience significantly elevated forces compared with the residual limb from increased reliance by the athlete. This may increase the risk for overuse injuries such as tendinopathies and stress fractures of the intact limb, as well as long-term degenerative changes such as osteoarthritis.^{173,203,207} Similarly, significantly altered use patterns and loads on the intact side in upper limb deficiency can result in pain and injury to either limb or axial structures. Overuse injuries such as shoulder impingement, rotator cuff tears, epicondylitis, and peripheral nerve entrapments are common in the intact limb of upper limb amputees.^{32,49,114,174} Differences in weight and swing excursion between the upper limbs, as well as the need to compensate at the shoulder for lost distal joint function, may cause significant asymmetry in the demands of the thoracic and cervical spines and paraspinal musculature and the periscapular stabilizing musculature, resulting in pain and dysfunction.^{10,174}

Injuries and Complications by Cause of Disability

Adaptive athletes are subject to many of the same injuries and complications from physical exertion as the able-bodied population. In addition, some specific diagnoses that result in disability also lead to increased risk of particular injuries and complications. There is certainly overlap between the categories of diagnoses resulting in disability. For example, many patients with SCI are wheelchair users; however, athletes with limb deficiency may compete seated or standing depending on the level of deficiency. For the purposes of this discussion, the injuries and complications discussed will be broken down into the wheelchair athlete and the limb-deficient athlete.

Wheelchair Athlete

Improvements in societal mindset, accessibility, and wheelchair technology have greatly increased the breadth and quality of athletic opportunities for the wheelchair user. Sports participation by wheelchair users has been found to be associated with improved functional outcomes, fewer physician visits, fewer hospitalizations, and fewer medical complications, while serving as a powerful means for improving inclusion and equality within society.^{48,50,51,83,113} Most athletes in this category have SCI, but eligibility may also extend to individuals with multiple amputations or neurological disorders such as polio, spina bifida, and cerebral palsy.^{50,51} Most of the discussion on wheelchair athletes will focus on considerations for the participation of athletes with SCI; however, these considerations may be generalized to other wheelchair athletes, depending on the amount of trunk and upper extremity function, level of sensation, and preservation of bowel and bladder function.

Given the degree to which wheelchair users rely on their upper limbs for mobility and activities of daily living, the importance of

recognizing, treating, and preventing upper limb injuries is magnified when compared with the able-bodied population. Because relative rest of the upper limb may be impossible or nearly impossible in this population, other measures should be considered. They may include splinting and orthotic prescriptions, admission to an inpatient setting if rest is required, home modifications, or additional assistance (e.g., Paralyzed Veterans of America).^{50,51}

Shoulder injuries. Wheelchair users are at increased risk for shoulder pathology, including pain, rotator cuff injuries, subacromial bursitis, acromioclavicular joint abnormalities, coracoacromial ligament thickening, subacromial spurs, distal clavicle osteolysis, and impingement syndrome.^{12,25,50,51,78} There are several factors contributing to the increased risk in this population, including repetitive motion, increased pressure in the shoulder joint during wheelchair propulsion, and muscle imbalances in the shoulder girdle caused by weakness.^{12,62,210,233} Shoulder muscle imbalance, with comparative weakness of the humeral head depressors (rotators and adductors), may be a factor in the development and perpetuation of rotator cuff impingement syndrome in wheelchair athletes.²⁷ In wheelchair users, upper limb injuries, including shoulder injuries, are particularly disabling because these patients rely on their upper limbs for weight bearing, transfers, and ambulation in addition to all the demands placed on the upper limbs in the able-bodied population. Despite increases in repetitive use and high-intensity activity, wheelchair athletes do not have a higher incidence of shoulder pain than nonathletic wheelchair users.^{50,51} In fact, participation in athletic competition appears to be protective.⁸¹ This is likely because of increased strength and endurance in the athletic population. Shoulder complaints among wheelchair users can be reduced by appropriate wheelchair design and the use of ideal propulsion techniques.

Elbow injuries. The elbow is another common site of both acute and chronic injuries in wheelchair athletes. The most common sources of elbow pain are lateral epicondylitis, osteoarthritis, and olecranon bursitis.^{50,51,177} Upper extremity peripheral nerve entrapments are frequently encountered in wheelchair athletes, with the most frequently reported elbow-related injury being ulnar neuritis.²⁸ Wheelchair users are at increased risk for ulnar neuropathy at the elbow.⁹¹ There is no evidence to suggest that wheelchair athletes are at a greater risk than nonathlete wheelchair users. Treatment of elbow pain and injuries in wheelchair users must be tailored to the individual needs of the patient, keeping in mind that many wheelchair users will be nonmobile if the individual is required to restrict weight bearing or otherwise limit activity involving their upper limb.^{50,51,78}

Wrist injuries. The wrist is another frequent site of both acute and chronic injuries in wheelchair athletes. The most common sources of wrist symptoms are carpal tunnel syndrome (CTS), ulnar nerve entrapment in Guyon canal, osteoarthritis, tendinitis, and de Quervain tenosynovitis.⁴ The carpal tunnel is the most common site of nerve entrapment in persons both able-bodied and disabled. Long-term wheelchair users have a 49% to 73% prevalence of CTS.²⁵¹ Wheelchair users with symptoms and physical examination findings of CTS have lower functional status compared with wheelchair users without CTS.²⁵¹

Upper limb fractures. The incidence of upper limb fractures in wheelchair athletes or wheelchair users is not known. Wheelchair athletes may be at greater risk for upper limb fractures caused by repetitive falls associated with many wheelchair sports, propulsion requiring positioning the hand in a location that is susceptible to injury from nearby wheelchairs or collisions, and relatively high speeds achieved during certain wheelchair sports.

Fractures should be treated as in the able-bodied population. Restricted upper limb weight bearing in a wheelchair user may result in a significant impact on ambulatory status.^{50,51,78}

Thermoregulation. After SCI, there is disruption of neuroregulatory systems that are involved in control of body temperature. Below the level of the lesion, athletes with SCI have impaired shivering (to produce heat) and impaired sweating and vasodilation (to dissipate heat). Athletes with tetraplegia are at increased risk compared with athletes with paraplegia.¹⁹² It is expected that paraplegic and tetraplegic athletes will have greater increases in body temperature with exertion and greater decreases in temperature with exposure to cold weather. Prevention of temperature-related injuries requires heightened awareness and monitoring and the use of appropriate clothing and equipment, availability of rehydration, and avoidance of extremes of temperature when possible. Frostbite is of particular concern during cold weather events. Athletes with SCI have impaired sensation and require frequent visual monitoring to prevent cold injuries.^{50,51,78}

Autonomic dysreflexia. Autonomic dysreflexia (AD) is a condition occurring from sympathetic outflow in response to a noxious stimulus being unregulated because of interruption of neural pathways after SCI, which at the level of T6 and above are at risk for AD. Symptoms of AD include paroxysmal hypertension, bradycardia, facial flushing, and headache. If the athlete with SCI develops hypertension caused by AD, which continues to increase without treatment, stroke or death may occur.¹⁸⁰ For acute blood pressure control, chewable nifedipine or topical nitroglycerin paste can be used. It is imperative to immediately assess for the noxious stimulus. The most common noxious stimuli that lead to AD include tight clothing, urinary or fecal retention, renal or bladder stones, pressure ulcers, infections, or intraabdominal pathology (e.g., appendicitis). Treatment involves sitting the patient upright with loosened clothing and is directed at alleviating the source of the AD after identification of the noxious stimulus.^{50,51,78} Boosting is the practice of intentionally inducing AD by athletes with SCI at or above the level of T6 for the purpose of improving athletic performance.^{19,99} Boosting has been shown to confer up to a 9.7% improvement in race time. In addition, to compete in a hazardous dysreflexic state, whether intentional or unintentional, is an extreme health risk. This dangerous practice should be discouraged and may be life threatening. For these reasons, the IPC strictly bans the practice of boosting, and it has developed a protocol to test for it. The IPC has also been working on the development of biomarkers that would be able to identify athletes who have been boosting. Blauwet et al. reviewed the current testing policy and practices that were used at three major international Paralympic events.¹⁹ Key parameters included the athlete's demographics, classification, and blood pressure measurements. An extremely elevated blood pressure was considered to be a proxy marker for AD, and a systolic blood pressure of at least 180 mm Hg was considered a positive test result. Seventy-eight tests for the presence of AD were performed during the three games combined; however, at this point, no athlete has tested positive with this testing method.¹⁹

Skin breakdown. After SCI, the patient will lose sensation below the level of the injury. This area of insensate skin leads to a risk of skin breakdown or pressure ulcer. Regardless of the level of the injury, the skin over the sacrum, coccyx, and ischial tuberosities is frequently insensate. These areas represent the highest risk for skin breakdown in athletes with SCI. Specific athletic events may result in increased risk of skin breakdown in additional areas. For example, wheelchair racers may have increased

risk of skin breakdown if the medial surface of the arm and forearm rubs against the wheelchair during propulsion. This may require customized equipment and padding to prevent skin breakdown in activity-specific, high-risk areas. Athletic wheelchairs commonly sacrifice pressure relief for higher performance. This should cause the athlete and caregivers to increase vigilance in monitoring for skin breakdown, changing position frequently, and limiting time in the wheelchair. This is particularly important when an athlete is transitioning to a new piece of equipment (e.g., a new wheelchair). Prevention is of paramount importance in this population.^{50,51,78}

At the first sign of skin breakdown or pressure ulcer, weight-bearing and athletic activities should be modified or restricted to prevent further injury. Pressure ulcers can be a significant cause of morbidity and mortality in persons with SCI regardless of athletic ability.^{50,51,78}

Heterotopic ossification. The development of ectopic bone formation (heterotopic ossification [HO]), in soft tissues surrounding major joints can occur after traumatic brain injury, SCI, burns, and total joint arthroplasty and even in the residual limbs of traumatic amputees.¹⁹⁰ The most common site of HO after SCI is the hip, but the knee, elbow, and shoulder may also be affected depending on the level of injury. In contrast, after amputation, HO occurs in injured tissues in the residual limb and may not be in the vicinity of a joint. In the wheelchair athlete with SCI, the initial presentation may mimic deep venous thrombosis or joint infection. Diagnostic testing is often required to exclude other diagnoses. Diagnosis of HO may be made by x-rays, bone scan, or trends in alkaline phosphatase levels. HO in residual limbs may increase the risk of skin breakdown or cause pain with weight bearing, edema, fever, restricted ROM, and activity limitations.^{50,51,78}

HO typically develops within the first 6 to 12 months after amputation during the time of the amputee's prosthetic training. Prevention of HO involves frequent ROM exercises. Given that most amputees are still learning how to use their prosthesis when HO typically develops, most HO cases are known before the amputee engages in athletic competition. It would be rare for development of new HO during competition. Recognition of HO in residual limbs allows for modifications in the design of a prosthetic socket to accommodate for the ectopic bone. Furthermore, monitoring for skin breakdown should be increased in amputees with HO. Treatment of HO is no different in the athlete than in the nonathlete.

Spasticity. Spasticity is a velocity-dependent increase in muscle tone that occurs after injury to the upper motor neuron. It is a common complication of people with upper motor neuron injuries, such as SCI, traumatic brain injury, and stroke. Spasticity may limit athletic participation by interfering with voluntary movements and restricting ROM. An increase in spasticity may be an indicator of a systemic or otherwise asymptomatic condition. For example, infections, intraabdominal pathology (e.g., appendicitis), skin breakdown, or bladder distention may have few symptoms that are sensed by a patient with SCI. Therefore a sudden increase in spasticity should lead to a search for underlying pathology. Treatment for spasticity consists of oral medications, including baclofen, dantrolene, tizanidine, and benzodiazepines; injectable medications such as botulinum toxin; and intrathecal medications such as baclofen. If spasticity is resistant to conservative treatment, surgery for tendon lengthening may improve hygiene, activities of daily living, and functional activities including participation in athletics.

Osteoporosis. Osteoporosis is a nearly universal complication of SCI. Due to decreased weight bearing, these athletes are predisposed to osteoporosis; nevertheless, many risk factors are independent of alterations in weight bearing. They include severity of the injury, spasticity, and time since injury.¹¹¹ Osteoporosis results in increased fracture risk of athletes with SCI. Because of impaired sensation below the level of the injury, athletes with SCI may not immediately complain of pain after a fracture. Thus it is imperative for the physician to be aware of other warning signs, which may include increases in spasticity or AD. Prevention of osteoporosis should include calcium and vitamin D supplementation for all athletes with SCI. Bisphosphonates may also be used for prevention.^{50,51,78}

Orthostatic hypotension. Orthostatic hypotension is a frequent complication in many individuals with SCI. Orthostatic hypotension occurs after SCI because of decreased sympathetic efferent activity in vasculature below the level of the injury and because of decreased reflex vasoconstriction. The result is venous pooling in dependent areas (lower limbs or abdomen) that occur with changes in position.¹²⁹ Symptoms include lightheadedness and dizziness, and syncope may occur if uncorrected. Prevention includes the use of lower limb compression stockings and abdominal binders, maintenance of hydration, and salt supplementation. If these measures are insufficient, pharmacological treatment with midodrine, fludrocortisone, or ephedrine may be helpful.¹²⁹ Each of these medications is considered a stimulant and is currently banned by WADA and the US Anti-Doping Agency as performance-enhancing drugs.²⁴⁹ Therefore in athletes with SCI who experience orthostatic hypotension, nonpharmacological prevention should be attempted before the use of pharmacological agents is considered.^{50,51,78}

Acute mountain sickness. Acute mountain sickness (AMS) is thought to be caused by alterations in the blood-brain barrier and cerebral vasculature that occur at high altitudes.⁶⁰ AMS involves a constellation of symptoms, including headache, nausea, weakness, and shortness of breath, and occurs with exposure to high altitudes. The incidence of AMS increases with increasing altitude, and hence there is an increased incidence in many winter sports because of competition at high altitudes. Given their altered neurophysiology and anatomy, athletes with SCI may be at higher risk of AMS.⁶⁰ Acetazolamide may be used as prophylaxis in high-risk scenarios. Treatment may include return to low altitude, acetazolamide, or dexamethasone.^{50,51,78}

Limb-Deficient Athlete

Individuals with partial or full loss of limbs are eligible to compete under the classification of amputee athlete. Limb loss may be congenital or traumatic and involve the upper limbs, lower limbs, or both. Depending on the specific sport, the limb-deficient athlete may compete as a standing or seated competitor. Given the significant variance in the location and length of the amputated limb, the functionality of the participant also has significant variance. Medical conditions and injuries that are unique to limb-deficient athletes are discussed later.

Skin disorders. After amputation of the limb and subsequent prosthetic fitting, the skin of the distal portion of the residual limb takes on the new role of being a weight-bearing surface. As a result, the distal residual limb is at increased risk for skin breakdown and other skin disorders. Common skin problems associated with prosthetic use include ulcers, inclusion cysts, calluses, contact dermatitis, hyperhidrosis, verrucous hyperplasia, lichenification, and infections.^{158,204} Rashes are commonly encountered

in persons using a prosthetic device and are frequently seen in amputee athletes because of issues related to increased perspiration. Bacteria, fungi, allergic reaction, or chemicals on the skin can form a rash. General principles of treating a rash include cleaning the residual limb more often and using a hypoallergenic soap and decreasing prosthesis use until the rash resolves. For bacterial or fungal rashes, antibacterial or antifungal medication is usually required. The rash may also be caused by an allergic reaction and may improve with the use of a mild corticosteroid cream. Such a rash should spur investigation as to the type of liner used with consideration of switching to a less irritating material or one that will allow greater perfusion of sweat away from the skin. Rash can be prevented by cleaning the residual limb and prosthesis as indicated.^{50,78} All persons with limb deficiency will have certain adaptations to skin changes with the residual limb regardless of whether they use a prosthesis for sports, and skin breakdown is always a concern. Currently, there is no literature that the effect of traditional preventative remedies for skin breakdown, such as additional liners, padded sleeves, or socks, has any consequences on athletic performance. It is speculated that skin breakdown may be more prevalent in athletes in competition secondary to increased perspiration. Some athletes may prophylactically treat their residual limb with an antiperspirant to decrease the amount of perspiration. Environmental conditions may also play a role, particularly for marine sports (swimming, kayaking, and rowing) and participation in hot weather. In amputee athletes, increased frequency and intensity of weight bearing associated with athletic training and competition also increase the risk of skin breakdown.

Identification of the functional demands the activity places on the residual limb will help to determine the design of the prosthetic socket. Some sports, such as running, require quick movement and many steps for a limited amount of time, whereas other sports, such as golf, require many more steps over a period of several hours. Both running and golf could be considered high-activity levels; however, there is a significant difference between the duration of these activities. Therefore it is imperative to understand the functional demands of each sport. The runner's prosthesis must absorb the impact of loading response, support body weight through midstance to allow a long stride length on the sound side, and provide a measure of propulsive thrust at the end of stance. In contrast, the golfer using a prosthesis endures long periods of standing, maintains overall stability when twisting during swings, and ambulates safely over uneven terrain, slopes, and inclines. Overall, the best treatment of dermatological issues is prevention, which can be achieved through education, close monitoring of prosthesis fit, strategic timing of donning a prosthesis, considerations of the time out of the prosthesis and liner, and responding to environmental factors.

Neuroma. A neuroma occurs at the distal end of a transected nerve in the residual limb of an amputee. When a neuroma is exposed to pressure, it creates paresthesias, dysesthesias, and radiating pain in the phantom distribution of the transected nerve. When a neuroma occurs at or near a weight-bearing structure, it can create severe pain with ambulation and weight bearing, limiting an athlete's ability to train and compete. Treatment may involve prosthetic modifications to relieve pressure on the neuroma, oral medications including antiepileptics and tricyclic antidepressants, injection of corticosteroids and local anesthetic into the neuroma, and radiofrequency ablation.¹²⁷ Unfortunately, many of the medications commonly used to treat neuromas may be restricted in competition by WADA; thus it behooves the sports medicine physician to be aware of the list of restricted

medications (Global Drug Reference at www.globaldro.com), which is modified annually.

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